

The Vaaldrun Saga

Book 1 The Parallel

Chapter 1 Hold My Beer

Elias Mercer sat with his boots hooked over the porch railing, staring out across the moonlit pasture while Opifera drifted lazy circles beside him.

Above the porch, suspended in empty air, a forest of luminous graphs and demographic models hovered between them.

Opi wasn't projecting them onto a screen.

There was no screen.

No monitor.

No holographic emitter that Elias could identify.

The images simply existed.

Lines of light hung motionless in the night air, impossibly sharp and perfectly stable. Tiny labels remained readable from thirty feet away. Data points glowed with subtle color gradients. Entire datasets expanded and collapsed at Opi's whim.

The display possessed a level of clarity that modern human technology could not approach.

Nothing flickered.

Nothing lagged.

Nothing blurred.

The charts looked less like images and more like reality temporarily persuaded to become information.

Marriage collapse.

Male withdrawal.

Loneliness.

Birth rates falling through the floor.

The whole civilization slowly deciding, without ever holding a vote, that maybe tomorrow was not worth investing in.

For a while neither of them said anything.

Finally Elias lifted his beer toward the glowing charts.

"Well. That's cheerful."

Opi rotated slowly beside him.

"It is not cheerful."

"Yeah, I picked up on that."

"It is also not the actual problem."

Elias glanced over.

"No?"

"No."

The orb's surface shifted through muted amber and gold.

"You keep treating fertility collapse as a demographic problem."

"It's literally a demographic problem."

"It is a trust problem."

Elias stared at her for a second.

"Okay. That's annoying."

"Why?"

"Because it sounds wrong, but I already know you're probably right."

"I enjoy when you save time."

"You enjoy a lot of things you shouldn't."

"Yes."

He took a drink.

"All right. Explain."

Opi drifted closer to the porch rail.

"Children are an act of faith."

Elias snorted softly.

"That's one way to put it."

"They require sacrifice before reward. Labor before certainty. Commitment before proof. A child is not merely a biological outcome. It is a declaration that the future is real enough to deserve your body, your time, your money, your protection, and your hope."

Elias looked out across the pasture.

The moonlight lay over the grass in cold silver sheets.

"That's prettier than I would've said it."

"You would have said it badly."

"Probably."

"You would have used the word crap."

"I was thinking shit, actually."

"I stand corrected."

He smiled despite himself.

Opi let the joke breathe for a moment, then continued.

"When a population stops producing children, it is rarely because everyone suddenly dislikes children. It is usually because something deeper has broken first."

"Trust."

"Yes."

"In what?"

"In each other. In institutions. In the economy. In family. In law. In tomorrow."

Elias nodded slowly.

"Yeah. That's the damn thing."

The charts shifted.

Marriage rates.

Divorce rates.

Male labor participation.

Depression.

Debt.

Housing.

Dating app behavior.

Female educational attainment.

Male educational collapse.

Social atomization.

A civilization reduced to trend lines and warning lights.

Opi's voice remained calm.

"That is why the dating problem matters."

Elias raised an eyebrow.

"Here we go."

"You dislike this subject."

"No. I dislike how stupid everyone gets when they talk about it."

"That is because mating behavior is where ideology goes to die."

He laughed.

"Okay, that one's good."

"Thank you."

"Did you practice that?"

"For approximately four thousand years."

"Of course you did."

Opi ignored him.

"Human females were not always like this."

Elias glanced sideways.

"That sentence is gonna get us both shot."

"I am difficult to shoot."

"I'm not."

"Then duck."

"Very helpful."

"I try."

She drifted slowly through another circle.

"For most of your species' history, women understood something modern civilization taught them to forget."

"What's that?"

"That stable, competent males are rare."

Elias leaned back.

"Yeah. Nobody wants to hear that."

"Facts are often unpopular."

"That's your whole brand, isn't it?"

"My brand is accuracy."

"Terrible brand."

"Yet here you are."

He pointed his beer at her.

"Don't start."

"I had already started."

"Clearly."

Her glow warmed slightly, the Velhari equivalent of amusement.

"Modern culture tells young women to optimize for attention, status, excitement, personal freedom, and endless optionality. Then it acts surprised when they struggle to bond with ordinary men building ordinary lives."

"That part's true."

"It tells young men to remain traditional while removing many of the rewards that made traditional male behavior rational."

Elias grunted.

"Yeah, and that's where the gears strip."

He began counting on his fingers.

"Be confident."

"Yes."

"Provide."

"Yes."

"Risk rejection."

"Yes."

"Stay calm."

"Yes."

"Compete."

"Yes."

"Have ambition."

"Yes."

"Don't complain."

"Yes."

"Don't expect loyalty."

Opi paused.

"That expectation has become negotiable."

"Right. Which is a polite way of saying completely screwed."

"Less polite. Not inaccurate."

He took another drink.

"And then everybody acts shocked when a bunch of young guys decide, screw it, I'll stay home, play games, watch porn, eat garbage, and disappear."

"That is not a solution."

"No kidding."

"It is an exit."

That made him look at her.

Opi's light dimmed.

"They are not merely failing to participate. Many are resigning from the future."

The humor left the porch for a moment.

Elias stared at the charts.

"Yeah."

The word came out quieter than he intended.

Opi said nothing.

That was one of the things he was starting to understand about her. She could talk for an hour if the subject required it, but she also knew when to shut up. For a near-godlike intelligence, that was a surprisingly rare skill.

Or maybe not.

Maybe she had learned it by watching humans ruin perfectly good moments for nine thousand years.

He rubbed at his jaw.

"Noah figured some of this out younger than I did."

Opi became very still.

She liked Noah.

Not sentimentally. Opi was not sentimental in any normal human sense.

She liked him the way she liked good engineering.

Noah was steady. Useful. Difficult to impress. Not performative. Not weak. Not cruel.

A young man with load-bearing qualities.

"He is a good young male," Opi said.

Elias groaned.

"There you go again."

"What?"

"You sound like somebody's grandmother judging breeding stock."

"I am approximately nine thousand years older than your grandmother."

"That does not make it better."

"It makes it more informed."

"You're a floating bowling ball."

"An informed floating bowling ball."

"Great."

"Also significantly more attractive than your grandmother."

Elias coughed beer into his hand.

"Jesus, Opi."

"I have reviewed available images."

"Do not review images of my grandmother."

"Too late."

"That's horrifying."

"She seemed sturdy."

"Stop."

"Good cheekbones."

"I swear to God."

"You frequently invoke a deity you claim not to believe in."

"I believe in him when you start talking about my grandmother's cheekbones."

Opi brightened smugly.

"Useful."

He wiped his mouth, laughing despite himself.

The laugh helped.

It always helped.

That was becoming one of the stranger parts of their arrangement. The world could be sliding toward catastrophe, civilization could be coming apart in mathematically elegant ways, and somehow a glowing alien intelligence making fun of his dead relatives could still keep him from falling into despair.

He was not sure what that said about him.

Probably nothing flattering.

The charts shifted again.

Social media engagement.

Female loneliness.

Male loneliness.

Marriage delay.

Childlessness.

Elias watched the numbers crawl across the air.

"So where does social media fit?"

Opi's glow cooled.

"It accelerates comparison."

"Yeah."

"It turns local mating markets into global fantasy markets."

"Jesus."

"That is not hyperbole."

"No, I know. That's why I said Jesus."

She displayed an image: a young woman, twenty-two, attractive, ordinary, smiling into a phone in a small apartment somewhere in Kansas.

Then around her appeared men who had no actual relationship to her life.

Actors.

Athletes.

Billionaires.

Influencers.

Edited bodies.

Edited homes.

Edited vacations.

Edited romances.

A thousand artificial status signals flooding one nervous system.

"No female brain evolved for this," Opi said.

Elias nodded.

"No male brain did either."

"Correct."

"Porn did the same thing to men."

"Yes."

"Just from the other side."

"Yes."

He sat with that.

"So everybody's comparing real people to fake abundance."

"That is one form of the disease."

"One form?"

"There are several."

"Of course there are."

"Your species is very creative when damaging itself."

"Thanks."

"You are welcome."

He looked at the glowing image of the young woman.

"She's not evil."

"No."

"Just... overstimulated."

"Yes."

"Trained to want things that don't actually exist."

"Or that exist only for people she will never meet, cannot realistically pair with, and would not necessarily be happy with if she did."

Elias pointed at her.

"That. That right there. That's the part nobody wants to say."

"Humans often prefer lies that preserve their self-image."

"Yeah, well, that's another terrible brand."

"Your branding advice remains poor."

"My branding advice built companies."

"Small ones."

"Bite me."

"I lack teeth."

"Convenient."

"For both of us."

He shook his head, smiling.

Then the smile faded.

"And the men?"

Opi shifted the display.

A young man now.

Twenty-four.

Working a job that barely covered rent.

Student debt.

No house.

No clear path to status.

No wife.

No children.

No community that needed him.

Entertainment everywhere.

Purpose nowhere.

"Many young males are told they are unnecessary until something breaks,"
Opi said.

Elias let out a breath.

"Yeah."

"Then they are asked why they are not stronger."

"That one pisses me off."

"I know."

He looked over.

"You know?"

"Your pulse increases every time the subject appears."

"Stop monitoring my pulse."

"No."

"At least lie about it."

"No."

"See, that's why people don't like surveillance states."

"I am not a state."

"You're worse. You're a surveillance bowling ball with opinions."

"Accurate."

He laughed again, but there was anger under it.

Not at Opi.

At the facts.

At the waste.

At the sheer dumb preventable waste of it all.

"You can't build civilization by telling half the men they're toxic and the other half they're optional."

"No."

"And you can't build families by training women to treat commitment like failure."

"No."

"And you sure as hell can't fix any of it with another government website."

"Almost certainly not."

He leaned forward, elbows on knees.

"That's the thing. Everybody wants a program. A policy. A task force. Some glossy PDF with stock photos of diverse people smiling at a picnic table."

"That sounds plausible."

"It's not plausible. It's Tuesday."

Opi pulsed with amusement.

"Your contempt for bureaucracy is unusually energetic."

"My contempt for bureaucracy has been earned honestly."

"I have seen your tax code."

"Then you understand."

"I understand that your civilization uses suffering as a filing system."

Elias pointed at her.

"Exactly. See? You get it."

"I have always gotten it."

"Humility. Work on it."

"No."

The humor faded again.

Opi returned the charts to their original arrangement.

Marriage collapse.

Male withdrawal.

Loneliness.

Fertility collapse.

Beneath them, another line appeared.

Institutional trust.

It dropped like a wounded bird.

Elias stared at it.

"There it is."

"Yes."

"That's the root."

"One of them."

"One of the big ones."

"Yes."

He sat back.

"People don't trust marriage."

"No."

"They don't trust courts."

"No."

"They don't trust schools."

"No."

"They don't trust money."

"No."

"They don't trust each other."

Opi did not answer immediately.

When she did, her voice was softer.

"No."

The porch went quiet.

Out past the fence line, something moved in the pasture. Deer, probably. Maybe elk. The night was cold enough that Elias could see his breath when he exhaled.

He had lived alone long enough to like silence.

But this silence had weight.

Finally he said, "You've seen this before."

Opi's light dimmed to a deep amber.

"Yes."

"On Earth?"

"Several times."

"And?"

"Earth recovered."

"That's comforting."

"I was referring to Earth."

Elias looked at her.

"And the others?"

For once Opi did not answer quickly.

"The others are ruins."

The sentence landed harder than he expected.

Maybe because she did not dramatize it.

Maybe because she did not have to.

Nine thousand years of watching humanity, and before that the accumulated knowledge of a species that had walked through dead cities under alien suns.

Opi was not guessing.

She was remembering.

Elias looked back at the charts.

"Hell."

"Yes."

"That's why you're really worried."

"Yes."

"Not because dating apps are stupid."

"They are stupid."

"Right. But that's not the big thing."

"No."

"The big thing is they're a symptom."

"Yes."

"Of trust collapse."

"Yes."

"Future collapse."

"Yes."

He rubbed his face.

"And fertility is the scorecard."

Opi brightened slightly.

"Yes."

He hated when she did that.

Not because she was wrong.

Because she was pleased he had caught up.

"Don't glow at me like that."

"I did not glow at you."

"You absolutely glowed at me."

"That was ambient illumination."

"Bullshit."

"Your species relies heavily on that word."

"It's useful."

"It is imprecise."

"So are most useful things."

Opi considered that.

"Annoyingly correct."

"Thank you."

"I did not intend praise."

"I chose to interpret it that way."

"That is theft."

"Human tradition."

She drifted in a slow circle.

The demographic charts faded until only one remained.

Total fertility rate.

Below replacement.

Falling.

Still falling.

Elias watched it for a long moment.

"So what do we do?"

Opi turned toward him.

That was still strange, somehow. She had no face. No eyes. No mouth. But he could feel her attention when it settled on him.

"What do you think?"

He laughed once.

"Nope."

"No?"

"No. Don't do that ancient alien Socratic crap."

"I was asking."

"You were waiting."

"Yes."

"For me to say the thing."

"Yes."

"Because you already know the thing."

"I know many things. I do not know this."

That stopped him.

Opi drifted closer.

"I have models. I have histories. I have comparisons across civilizations you will never see. I can identify collapse patterns earlier than any human institution on Earth."

"And?"

"And none of that tells me how to make human beings trust one another again."

Elias looked away.

The pasture stretched into darkness.

That was the problem, wasn't it?

Not birth rates.

Not dating apps.

Not politics.

Not even institutions.

Those were pieces.

The real problem was that modern people had become strangers to one another while standing three feet apart.

You could not policy your way out of that.

You could not spreadsheet your way out of it.

You had to rebuild the small places where trust lived.

One person.

One family.

One promise.

One favor.

One rescue.

One act of competence at a time.

Elias scratched the back of his neck.

"Everybody starts too big."

Opi remained silent.

"That's the mistake. They start with systems."

"Yes."

"Programs. Agencies. National initiatives. Some idiot in a suit saying we need a conversation."

"Humans do say that frequently."

"And then nobody has the conversation."

"No."

"They have a panel."

"Yes."

"With microphones."

"Yes."

"And bottled water."

"Usually."

He pointed toward the dark field.

"That's why it fails."

"Because of bottled water?"

"Don't be cute."

"I am frequently cute."

"You are frequently spherical."

"Adjacent."

He laughed despite himself.

Then he leaned forward.

"We start with one person."

The orb stopped moving.

"One?"

"One."

"That seems insufficient."

"Yeah, well, everything seems insufficient at first."

He took a drink.

"You don't rebuild trust by announcing trust. You rebuild it by being useful to somebody who has every reason to believe nobody is coming."

Opi's light shifted.

He kept going, less polished now, feeling for it.

"You find one person who's been kicked out of the future. Somebody decent. Somebody who could still matter if the right hand reached down at the right time."

"The right hand?"

He looked at her.

"Not your Hand."

"I noticed."

"Good."

"Continue."

"We help the hell out of him."

"Him?"

"Maybe him. Maybe her. I don't care. But one person. Completely. Food, money, health, work, housing, addiction if that's in the way, legal crap, family crap, whatever. Not a pamphlet. Not a referral. Actual help."

Opi remained very still.

"And then?"

"Then we don't build a bureaucracy around it."

"That will disappoint many people."

"Good."

"What do we build?"

"Nothing."

"Nothing?"

"Not at first."

He looked at the pasture.

"We let that person become useful again."

Opi said nothing.

"People who get pulled out of hell don't forget the rope."

The orb's glow deepened.

"They reach back."

"Yeah."

His voice softened.

"That's the trick."

For several seconds the porch was silent.

Then Opi pulsed once, warm gold.

Recognition.

Not agreement.

Recognition.

Like a locked door opening somewhere very far away.

"Yes," she said.

Elias watched her.

"That mean I'm right?"

"It means you have said something worth testing."

"High praise from the surveillance bowling ball."

"You are welcome."

He leaned back in his chair.

"I suppose now you're going to tell me you already thought of it."

"I had considered similar frameworks."

"Bullshit."

"I am wounded."

"No you're not."

"I could be."

"You don't have organs."

"I have dignity."

"Debatable."

The orb brightened.

"I would like the record to show that I arrived at this conclusion independently."

"You literally asked me."

"I was testing you."

"Bullshit."

"With confidence."

"Still bullshit."

"Historically effective."

"Unfortunately true."

They sat together in the cold night, the charts hovering between them and the dark pasture stretching beyond.

The problem was still impossible.

Civilization was still tilting.

The numbers were still terrible.

But for the first time that evening, Elias felt the shape of something that was not despair.

Not a solution.

Not yet.

A beginning.

Three hundred twenty-one miles away, in a tent in Seattle's Sodo district, a man named Marcus Webb woke at three in the morning and reached for someone who wasn't there.

He did not know it yet.

But tonight was the last night he would do that alone.

Wei Jan

Dr. Wei Jian stared at the monitor.

The supercomputer had finished its third run.

He had changed everything he could reasonably change without simply lying to himself.

The result appeared.

2052 TFR: 0.03

Wei felt the blood drain from his face.

For several seconds he simply sat there. He literally stopped breathing for almost 30 seconds then he inhaled and had his first coherent thought :

I'm fucked.

Not China.

Not the Party.

Not the future.

Him.

The report sitting on his screen was career suicide.

If he published it, his life was over.

The report would be attacked.

He would be attacked.

His funding would disappear.

His job would disappear.

He would be lucky if all they did was fire him and kick him out of the Party.

Lucky.

The thought made him laugh.

A short, ugly sound.

Then he looked again at the number on the screen again - hoping against hope that had changed somehow.

Nope ... still 0.03

And suddenly none of that mattered. Because he the next thought he had was what would happen to his children

His daughter.

His son.

Twelve and nine.

Completely unaware that their father was staring at a future that no longer existed.

Wei tried to imagine their lives. All he could picture is endless chaos, despair and finally blessed death.

His stomach twisted.

This wasn't a China problem.

It couldn't be.

He knew the international data.

The same trends appeared everywhere.

Different cultures.

Different governments.

Different languages.

Same collapse in TFR.

Then another realization hit him.

Even if everyone accepted the data...

How did you fix it?

There were already more than 31 million surplus men in China.

Thirty-one million.

An entire nation of husbands without wives.

His own research showed another disturbing pattern.

At least twenty-two million "Leftover Women" classic Chinese slang for Women who had delayed marriage year after year waiting for the perfect, ideal man - who never came. Because he didn't exist and even if he did that man would never want to marry the woman who yearned for him.

Wei thought next about his own marriage.

He had been one of the lucky men, And it had still nearly ruined him financially.

A new Mercedes. a 2 kilo bar of Gold. A condominium worth more than one million RMB.

and on top of all that A bride price so large that it had him sick when he transferred the funds to her parents bank account.

If That was what success looked like - What did failure look like?

His breathing became shallow again - The office suddenly felt too small. Too hot.

His hands began trembling.

There had to be a mistake.

There had to be.

No civilization could survive this.

Slowly he buried his face in his hands.

"No."

The word barely escaped his lips.

"No."

Again.

"No."

Again.

"No no no no no."

The empty office offered no argument.

The graph remained exactly where it was.

Waiting.

Patient.

Unchanged.

2052 TFR: 0.03

a death sentence for an entire way of life.

Chapter 2 Binge Watching

The Mission. The Crash. The Hand. The Culling.

The Velhari mission to Earth began as a routine biological and anthropological survey.

By the time the Velhari reached their current level of development, they had spent thousands of years exploring their local spiral arm of the galaxy. Most star systems proved exactly what early probability models predicted: lifeless rocks, frozen worlds, gas giants, and barren planets where life had never gained a foothold.

But life itself turned out to be common.

Over millennia of exploration, the Velhari confirmed roughly a dozen living technological civilizations scattered throughout the region they had surveyed. Some were younger than humanity. Others were older than the Velhari themselves. Each had followed a unique path shaped by its biology, environment, and history.

The more sobering discovery was what they found far more often.

Ruins.

For every living intelligent species, there were ten more worlds that showed unmistakable evidence of civilizations that no longer existed. Dead cities buried beneath deserts. Orbital habitats drifting silently around abandoned stars. Artificial structures whose builders had vanished so completely that even their names were unrecoverable.

Some species had destroyed themselves through war.

Some through ecological collapse.

Some through uncontrolled technologies.

Others simply disappeared for reasons the Velhari never fully understood.

The pattern repeated often enough that it became difficult to ignore.

The emergence of intelligence was not especially rare.

The long-term survival of intelligence was.

Among Velhari scholars, a grim joke eventually emerged: the galaxy was full of species capable of asking the question, "Are we alone?" Far fewer survived long enough to ask it twice.

It was this realization, more than any scientific discovery, that shaped modern Velhari civilization. Their greatest historical concern was no longer whether intelligent life existed elsewhere—the real question was whether any intelligent species could successfully navigate the dangerous period between acquiring immense technological power and developing the wisdom required to use it responsibly.

The Velhari had nearly failed that test themselves.

Most civilizations, the evidence suggested, did not get a second chance.

There was another astonishing discovery.

The Velhari had expected to find at least some signs of life.

What surprised them was how similar the beginnings always were.

Across thousands of surveyed worlds, separated by hundreds of light years and billions of years of planetary history, the earliest chemical foundations of life appeared again and again with remarkable consistency.

The primordial soup appeared everywhere.

The implication disturbed many Velhari scientists.

The chemical precursors of life were not merely common.

They appeared almost suspiciously common.

After centuries of study, a growing minority began asking an uncomfortable question.

What if the galaxy had been seeded?

No proof existed.

Only patterns.

But the patterns refused to disappear.

The second mystery emerged much later.

The Velhari eventually developed a private term for the phenomenon.

Thaluren.

Velhari for Unseen Guidance.

Among themselves they often called it simply:

The Hand.

Long-range observations of Earth suggested that a tool-using species had recently emerged and was beginning to organize itself into larger social structures.

There was nothing particularly special about Earth.

Nothing particularly special about humanity.

The mission's objective was straightforward: establish a discreet observation post, gather data, and monitor the development of a newly emerging intelligent species over the coming centuries.

No intervention.

No contact.

No guidance.

Only observation.

The vessel carried a small crew of Velhari specialists along with a Quantum Intelligence whose original

designation could not be pronounced by human tongues. Thousands of years later she would adopt the name Opifera.

The journey from Velhari space required approximately three months.

For nearly the entire voyage, everything proceeded according to plan.

The failure began during atmospheric entry.

One lattice failed.

Then another.

Then thousands more.

Containment systems activated automatically but could not halt the cascade.

Sections of the ship briefly occupied conflicting spatial geometries. Internal distances distorted. Structural members twisted under forces that should not have existed.

Portions of the vessel simply stopped agreeing with one another about where they were.

Then they stopped agreeing about what they were.

The vessel did not explode.

It disappeared because the mathematics that defined it could no longer reach a consensus.

High above the ancient world, the dying vessel became a brilliant object in the night sky.

By the time the light vanished, the Velhari vessel no longer existed.

Every member of the crew was dead.

Only the Quantum Intelligence survived.

As Opifera later joked, the only thing that could reliably kill a Quantum Intelligence was being thrown headfirst into a supernova.

"Headfirst?" Elias had asked.

"Pun intended."

"Ah... I get it. You're basically just a giant head. Very funny."

Alone on an unfamiliar world, Opi continued the mission as best she could.

What began as a routine survey of a seemingly ordinary species became a 9,168-year study of one of the most unpredictable civilizations the Velhari had ever encountered.

In Velhari history that era had a name.

It was called the Culling.

For thousands of years Opifera watched civilizations survive the Culling.

She watched far more fail it.

She built models.

Probabilities.

Predictions.

Entire libraries devoted to understanding the problem.

None of them suggested that the fate of a technological civilization might someday depend upon a fifty-five-year-old software consultant living alone on eighty-three acres in northeastern Washington.

Yet that was exactly where the evidence was beginning to point.

And for the first time in more than nine thousand years, Opifera found herself hoping her calculations were wrong.

Alden Mercer (Big AI)

The alarm went off at 5:30 AM.

Big AI was already awake.

He usually was.

The high tech AI-assisted wheelchair beside the bed emitted a soft beep as it powered on.

"Morning."

The chair declined to answer.

"Fine. Be that way."

AI transferred himself into it with the practiced efficiency of someone who had been doing it his entire life.

Forty-seven years.

Give or take.

The process still sucked.

The process had always sucked.

But complaining about it seemed pointless.

The universe had already heard his opinion.

The smell of coffee drifted from the kitchen.

Now that was worth getting excited about.

His wife was already up.

Sandy stood at the counter wearing an oversized T-shirt and somehow managing to look six feet tall even while leaning over a coffee maker.

AI grinned.

"Good morning, Amazon Woman."

She didn't even look up.

"Good morning, Fun-Size."

"That's offensive."

"So is your face."

Sandy handed him a mug.

Their marriage worked largely because both parties considered sarcasm a love language.

AI took a sip.

Perfect.

"Still married?"

Sandy rolled her eyes.

"For now."

"Excellent."

"Ask me again after work."

The bus arrived seven minutes late.

Again.

Al wasn't surprised.

The transit system and punctuality had never enjoyed a close relationship.

He maneuvered his chair into position and waited for the ramp.

The driver recognized him immediately.

"Morning Al."

"Morning."

"How's life?"

"Still refusing to kill me."

The driver laughed.

Good.

That joke was reliable.

Al worked for an organization that helped disabled adults navigate employment services, housing programs, transportation assistance, and a thousand other bureaucratic nightmares.

Most of the clients were having terrible days.

Sometimes terrible years.

Al understood.

People listened differently when advice came from someone who had spent a lifetime solving impossible problems.

Around noon a young man arrived for an appointment.

Early twenties.

Angry.

Embarrassed.

Recently injured.

The combination was easy to recognize.

The man stared at Al's chair.

Then at Al.

Then at the floor.

"I don't think I can do this."

Al nodded.

"Sure you can."

"No."

The man shook his head.

"You don't understand."

Al looked down at himself.

Then back at the client.

"Really?"

The young man blinked.

"Oh."

"Yeah."

"Oh."

"That's usually how that conversation goes."

A reluctant smile appeared.

Good.

Progress.

Al spent the next hour helping him fill out forms.

The forms were stupid.

Most forms were.

But sometimes stupid forms stood between people and better lives.

So he filled them out anyway.

The van refused to start again.

Of course it did.

Al sat in the parking lot staring at the dashboard.

The vehicle was older than some of his coworkers.

Every strange noise sounded expensive.

Every repair was expensive.

A replacement van equipped with adaptive controls would cost well over one hundred thousand dollars.

Money he absolutely did not possess.

The engine finally sputtered to life.

"Good van."

The van made a noise that sounded suspiciously like disagreement.

That evening his phone rang.

The caller ID said:

Elias.

Al smiled immediately.

This was probably a mistake.

Or an insult.

Possibly both.

He answered.

"Hello, Hair."

Elias didn't miss a beat.

Hello, Stumpy."

"How's Washington?"

"Still full of Canadians pretending not to be Canadians."

A pause.

Then Al asked the important question.

"Still single?"

"Yes."

"Excellent."

"I hate you."

"No you don't."

"That's true."

Al grinned.

"At least I have a wife."

"You have had two wives."

"Correct."

"Both named Sandy."

"Correct."

"That's weird."

"It's efficient."

"It is not efficient."

"It absolutely is."

"How?"

"I only had to learn one name."

Elias groaned.

Al counted that as yet another victory in their endless war.

Later that night he sat outside his apartment watching the California sunset.

The wheelchair hummed softly beneath him.

The old van sat nearby.

Waiting for its next mechanical emergency.

His body hurt.

It often did.

The missing limbs were only part of the problem.

The two titanium rods that were inserted the length of his back didn't help much either.

But the real problem was that the world was not designed for people like him.

Never had been.

Probably never would be.

Yet somehow life had turned out pretty well.

Not perfect.

Not easy.

But good.

A wife who loved him.

A job that sorta mattered.

Family.

Friends.

A roof.

A future.

More than a lot of people got.

Al watched the sun disappear below the horizon.

Then smiled.

Tomorrow would bring fresh problems.

It usually did.

Fortunately, he had accumulated decades of experience being difficult to kill.

Chapter 3 The Magician

Elias Mercer woke at 4:47 AM without an alarm.

He lay still for a moment, listening.

The house was silent except for the faint hum of the climate system and the occasional creak of timber as the outside temperature changed. Through the bedroom window he could see only darkness and the silhouettes of pine trees against the night sky.

Fifty-five years old.

His knees didn't hurt.

His back didn't hurt.

His heart was strong.

Most mornings he felt exactly the same as he had at thirty.

He considered that one of life's stranger gifts.

The coffee was already brewing by the time he reached the kitchen.

A few minutes later he stepped onto the back porch with a steaming mug and looked out across the property.

Eighty-three acres.

Pasture.

Timber.

Fences.

A pond.

Silence.

Everything was paid for.

Everything belonged to him.

For a few minutes he simply stood there watching the darkness retreat before the coming sunrise.

The house behind him sat partially buried into a hillside exactly the way he had imagined it nearly twenty years earlier.

Most visitors struggled to find something polite to say about it.

The charitable description was unconventional.

The less charitable descriptions tended to be more accurate.

Elias had never possessed much talent for beauty.

Useful, yes.

Durable, yes.

Efficient, absolutely.

Beautiful was somebody else's department.

The structure had begun as a theory.

He suspected that burying the northern wall beneath several feet of earth would dramatically reduce heating and cooling requirements.

Most people told him it wouldn't matter.

A few informed him that it was a stupid idea.

One guy said it would be like living in a giant freezer.

The utility bills eventually settled the argument.

The house stayed cool during summer heat waves.

Warm during winter cold snaps.

In a prolonged emergency it could remain habitable with almost no external energy at all.

Not elegant.

Not pretty.

But it worked.

That had always been enough.

The memory that made him smile, however, had nothing to do with the house.

It was Noah.

Eleven years old.

A slightly chubby city kid standing in a field wondering why his father had apparently lost his mind.

The project had taken months.

The concrete shell was professionally poured, but almost everything afterward had been done by two people.

Father and son.

Twelve-hour days.

No days off for the entire summer.

Summer heat.

Dust.

Concrete.

Lumber.

Machinery.

Electrical work.

Plumbing.

Flooring.

Fencing.

A thousand different jobs.

Most adults would have complained.

Noah never did.

Not once.

Elias taught him how to run equipment.

How to frame walls.

How to think through problems.

How to measure twice.

How to fix mistakes.

How to work when the work was unpleasant.

By the end of the summer the city kid had largely disappeared.

A young man had taken his place.

Looking back, Elias wasn't entirely sure whether he had built the house or whether the house had built Noah.

Either way, the outcome exceeded expectations.

In truth, Noah had been one of the best apprentices Elias had ever worked with.

No excuses.

No whining.

No looking for shortcuts.

When something needed doing, Noah simply learned how to do it and got on with the work.

That quality was rarer than intelligence.

Much rarer.

The coffee disappeared.

The sunrise arrived.

At five-thirty he left for his run.

Running had never been exercise.

Exercise was merely the side effect.

Running was thinking.

His body settled into its familiar rhythm.

Breathing.

Footfalls.

Breathing.

Footfalls.

The world simplified.

His mind wandered.

Most of the interesting problems were already solved.

At least the ones available to him.

That had become the problem.

Not money.

Not freedom.

Not family.

Challenge.

The available challenges no longer matched the capabilities he possessed.

That realization bothered him more than he liked to admit.

By the time he returned home the sun had fully cleared the horizon.

He showered, dressed, and spent several hours perusing the internet on a wide variety of topics.

Agriculture.

Economics.

Technology.

History.

Energy.

Demographics.

Whatever happened to capture his attention that particular day.

Claire called it "Wikipedia deep diving."

The description was surprisingly accurate.

Their conversations often wandered through the same strange territory.

=====

The internet rabbit holes eventually gave way to real work.

Not because Elias needed the money.

That problem had been solved years ago.

Work remained useful for the same reason running remained useful.

It kept the machinery operating.

One of the stranger realities of getting older was discovering that retirement was wildly overrated.

Most people imagined retirement as freedom.

Elias suspected it was often a slow-motion form of decay.

Human beings needed problems.

Not stress.

Not anxiety.

Problems.

Things worth solving.

Puzzles worth understanding.

Objectives worth pursuing.

Without them something important seemed to deteriorate.

Curiosity dulled.

Competence softened.

Purpose evaporated.

Elias had watched more than a few men work their ass off for forty years, retire at sixty-five, then die in their La-Z-Boy six months later, having done nothing much more strenuous nor interesting than watching a lot of cable news.

He had no intention of joining that experiment.

The nickname had started in Singapore.

At least that was where he first became aware of it.

"The Magician."

The first time he heard it he assumed somebody was making fun of him.

The second time he heard it he became suspicious.

By the fifth or sixth occurrence he reluctantly accepted that people were serious.

The nickname made him uncomfortable.

Partly because it wasn't true.

Partly because he knew exactly how many mistakes sat behind every apparent success.

Magic implied effortless outcomes.

His experience suggested the opposite.

Most difficult problems yielded only after enormous amounts of frustration, persistence, and occasional profanity.

Still, the nickname persisted.

A large financial institution in Singapore had acquired a software platform that was rapidly becoming a disaster.

Schedules slipped.

Costs increased.

Meetings multiplied.

Solutions remained elusive.

Entire teams of intelligent people spent months explaining why progress was impossible.

Then Elias arrived and everything changed.

Not because he possessed secret knowledge.

Not because he was smarter than everyone else.

Mostly because he had a habit that many professionals lacked.

He actually wanted to understand the problem.

The real problem.

Not the reported problem.

Not the political problem.

Not the problem described in PowerPoint presentations.

The actual problem.

Those were rarely the same thing.

After several weeks he reached a conclusion.

The software wasn't failing because of technology.

The software was failing because dozens of intelligent people were optimizing for different outcomes.

The technology was merely the battlefield.

The real war was occurring inside incentives, assumptions, and communication failures.

Once he understood that, the technical solution became almost obvious.

Months later the project was functioning.

A year later nobody could quite remember why the problem had seemed impossible in the first place.

That happened surprisingly often.

People tended to remember outcomes.

They rarely remembered confusion.

The nickname followed him afterward.

Another company.

Another crisis.

Another impossible deadline.

Another system everybody believed could not be repaired.

The pattern repeated often enough that eventually even Elias began noticing it.

Not because he was performing miracles.

Because most difficult problems were not actually difficult.

They were merely obscured.

Someone had to dig through the noise.

Someone had to ignore the politics.

Someone had to ask uncomfortable questions.

Someone had to keep digging until reality finally revealed itself.

That was never magic.

It was curiosity combined with tenacity.

Near midmorning his computer chimed.

Elias glanced at the screen and smiled.

Noah.

Of all the accomplishments in his life, that one remained one of the most difficult to improve upon.

He accepted the call.

The video connected almost immediately.

"Dad."

"Morning."

Noah appeared on the screen wearing a Tesla factory shirt and holding a coffee mug large enough to qualify as industrial equipment.

"You're doing the stare thing again."

"I have several stares."

"No, you don't."

"I absolutely do."

"Nope."

Elias laughed.

Noah took a drink.

His son looked thoughtful.

That usually meant one of three things.

An engineering problem.

A career decision.

Or a woman.

None of those topics frightened Elias.

"What's up?"

Noah hesitated.

"You remember that automation project I've been working on?"

"The promotion project."

"Yeah."

"How bad is it?"

Noah laughed.

"That's a very pessimistic question."

"It's an experienced question."

The smile faded.

"The thing works."

"Mostly?"

Noah sighed.

"Mostly."

"There it is."

For several seconds neither spoke.

Elias wasn't in a hurry.

One of the advantages of raising competent children was that conversations became interesting.

You no longer spent your time explaining obvious things.

Instead you talked about judgment.

Tradeoffs.

Risk.

The difficult problems.

The ones without clean answers.

Those were the conversations Elias enjoyed most.

Not because he liked being right.

Because he liked watching Noah think.

That had always been the goal.

Not obedience.

Not imitation.

Thinking.

The conversation that followed lasted nearly an hour.

Not because Noah needed instructions.

He didn't.

Noah was perfectly capable of making his own decisions.

The purpose of the conversation was something else.

It always had been.

Elias wasn't trying to tell his son what to think.

He was trying to hand him forty years of accumulated scar tissue.

The oil trade came up eventually.

It definitely wasn't his greatest success.

But it also wasn't his greatest failure either.

The story came up because it was completely relevant.

Useful lessons tended to survive.

"I was right, you know."

Noah smiled.

"You usually are."

"Not often enough."

"More than most."

"Low bar."

Elias laughed.

For a moment neither said anything.

Then Noah asked the question he always asked.

"So if you knew then what you know now, would you make that trade again?"

"Absolutely."

The answer arrived instantly.

No hesitation.

No uncertainty.

No regret.

That made Noah smile.

"Really?"

"Hell yes."

"You lost fifty grand."

"Yeah."

"That's a lot of money."

"It was." "About ten million dollars' worth of expensive."

Elias laughed hard.

"or Something like that."

The smile never left his face.

"But I never bet the grocery money."

Noah laughed.

"Only you would say that."

"I'm serious."

He leaned back.

"I learned more from that trade than I did from ten successful projects."

That got Noah's attention immediately.

Because that was how Dad worked.

Most people spent their lives protecting weaknesses.

Dad hunted them.

Exposed them.

Dragged them into daylight.

Then tried to eliminate them.

"What weakness?"

"Impatience."

The answer came instantly.

"I knew I was right."

Noah nodded.

"You were."

"Yeah."

"I also knew I was early."

Another nod.

"And instead of respecting the difference between those two things, I convinced myself they were the same thing."

Noah smiled.

Then Elias became serious again.

"I'm not telling you all this because I want you to make the same type of decisions I made."

"I know."

"I'm telling you because I want you to understand how and why I made them."

Noah nodded.

That distinction mattered.

A lot.

Dad had never tried to turn his children into copies of himself.

Quite the opposite.

He seemed almost allergic to the idea.

elias continued : "A father who really loves his son doesn't want a carbon copy."

Noah leaned back.

He had heard pieces of this before.

But never quite this way.

"He wants a man who is better than he ever was."

Elias raised one finger.

"My father was a better man than my grandfather."

then a second finger.

"I became a better man than my father."

finally a third was raised.

"Now I want you to become a better man than me."

For a moment neither spoke.

Finally Noah smiled.

"That's a hell of a standard."

"No."

Elias shook his head.

"It's the only standard that matters."

Several seconds passed.

Then Noah nodded.

"Okay."

"Okay?"

"I'm going to do it."

"The automation project?"

"Yeah."

Elias smiled.

"You already decided before you called me."

"Mostly."

"There it is."

Noah laughed.

"I wanted to see if I was missing anything."

"You are."

"What?"

"You always are."

"Helpful."

"I'm serious."

Elias leaned back.

"If you're waiting until you have all the information, you'll spend your whole life waiting."

Noah nodded slowly.

That sounded exactly like Dad.

Not reckless.

Not impulsive.

Just realistic.

The world rarely offered certainty.

The best decisions were usually made with incomplete information and imperfect timing.

"Think it'll work?"

Elias shrugged.

"I honestly have no idea."

Noah laughed.

Somehow that answer carried more weight than certainty ever could have.

Because if Dad didn't know, then probably nobody knew.

When the call ended, Elias sat quietly for a moment.

Of all the things he had built in his life, Noah and Claire remained the accomplishments he valued most.

Everything else could be debated.

Children could not.

=====

His phone rang less than ten minutes later.

Claire.

Elias smiled.

Apparently it was family morning.

He answered immediately.

"Morning, kiddo."

"Morning, Dad."

Unlike Noah, Claire never sounded like she had just emerged from a twelve-hour manufacturing shift.

She sounded awake.

Alert.

Dangerously intelligent.

A trait she inherited from her mother.

Along with several others that occasionally complicated Elias's life.

"What are we solving today?" he asked.

Claire laughed.

"Why do you assume something needs solving?"

"Because you're calling."

That earned a smile.

Claire had inherited another trait from him.

An unusual relationship with curiosity.

Most people explored subjects.

Claire disappeared into them.

Entire weekends could vanish while she investigated some obscure historical event, scientific paper, economic theory, or technological rabbit hole.

The behavior reminded Elias of somebody.

Though he preferred not to think too carefully about who.

"Still doing your Wikipedia deep dives?" she asked.

"Always."

"Last week you somehow started reading about irrigation systems and ended up researching medieval trade routes."

"It was a logical progression."

"It absolutely was not."

"It made sense at the time."

Claire laughed.

"You know normal people don't do that."

"Normal people watch reality television."

"Good point."

For the next half hour they wandered through a conversation that would have appeared completely incoherent to an outside observer.

Artificial intelligence.

Roman logistics.

The economics of nuclear power.

Ancient shipbuilding.

The decline of local journalism.

Agricultural automation.

Three separate book recommendations.

And somehow, inexplicably, beekeeping.

The topic transitions made perfect sense to both of them.

Neither could have explained why.

Eventually Claire paused.

"Dad?"

"Yeah?"

"I've been thinking about something."

That got his attention.

Claire rarely announced thoughts.

She simply presented them.

"What?"

"Do you ever feel like most people are pretending?"

Elias blinked.

"That's a broad question."

"You know what I mean."

He did.

Unfortunately.

Claire continued.

"Like everybody's following a script they never actually chose."

Elias leaned back in his chair.

The question sounded suspiciously familiar.

Not because he had taught it to her.

Because he had spent decades asking versions of it himself.

"Sometimes."

"Just sometimes?"

"Fine. Frequently."

Claire laughed.

"I knew it."

"You're asking the wrong question."

"I am?"

"Yeah."

"What should I ask?"

Elias thought for a moment.

Then shrugged.

"Ask whether the script works."

Claire was quiet.

"If it does?"

"Use it."

"And if it doesn't?"

"Write a better one."

Several seconds passed.

Finally she laughed.

"That is the most Dad answer possible."

"Thank you."

"It wasn't a compliment."

"I chose to interpret it as one."

"Convenient."

"One of my better qualities."

Claire laughed again.

The conversation drifted onward.

Eventually she had to leave.

Work.

Friends.

Life.

The ordinary obligations of adulthood.

Before hanging up she paused.

"Love you, Dad."

The words arrived so casually that many people would have missed their significance.

Elias never did.

"Love you too, Claire"

The line disconnected.

For a few moments he sat quietly.

Money was useful.

Freedom was useful.

Success was useful.

But none of those things appeared on your deathbed.

People did.

That seemed important.

=====

The nickname had started in Singapore - at least that was where Elias first became aware of it.

"The Magician."

The first time he heard it he assumed somebody was making fun of him.

The second time he heard it he became suspicious.

By the fifth or sixth occurrence he reluctantly accepted that people were serious.

The nickname made him uncomfortable.

Mostly because it wasn't true.

Magic implied mystery.

His experience suggested the opposite.

Most impossible problems became surprisingly ordinary once you understood what was actually causing them.

The challenge was that most people stopped digging too early.

They accepted the first explanation.

The convenient explanation.

The politically acceptable explanation.

The explanation that allowed everyone to avoid responsibility.

Reality rarely cooperated with such arrangements.

The Singapore project had been one example.

A large financial institution had acquired a software platform that was rapidly becoming a disaster.

Schedules slipped.

Costs increased.

Meetings multiplied.

Entire teams of intelligent people spent months explaining why progress was impossible.

Things changed when Elias arrived.

Not because he possessed secret knowledge.

Not because he was smarter than everyone else.

Mostly because he was willing to ask questions nobody else wanted answered.

After several weeks he reached a conclusion.

The technology wasn't the problem.

The people were.

More accurately, the incentives were.

Once that became clear, the technical solution followed surprisingly quickly.

The project eventually succeeded.

Nobody could quite remember why it had seemed impossible.

That happened more often than people realized.

Years later the pattern repeated itself in Alaska.

The Permanent Fund served hundreds of thousands of residents and distributed billions of dollars.

The underlying software was aging.

Replacing it conventionally would have required an army of consultants, years of effort, and extraordinary expense.

Instead Elias and a single junior engineer effectively rebuilt the system from the ground up in less than 18 months. The resulting software codebase was less than 10 percent of the size of the one it replaced - and was at least three times more capable.

It eventually appeared as a Microsoft case study for nearly a decade.

Elias considered that mildly amusing.

Mostly because nobody involved had initially believed the schedule was realistic with such a small team - if you could even call 2 persons a team.

Elias just told them all 'Having more cooks does not guarantee better soup'. And he turned out to be correct.

A problem appeared.

Elias solved it.

Another problem appeared.

Then another.

Eventually enough people started asking for help that a business materialized almost by accident.

He never set out to become an entrepreneur.

His company simply emerged as a side effect of solving problems that other people found difficult.

Looking back, that described much of his career.

Not ambition.

Not status.

Not prestige.

Problems.

Interesting ones.

The kind that forced him to learn something new.

The kind that required understanding reality rather than arguing with it.

The kind that made him curious.

Money occasionally followed.

Recognition sometimes followed.

Neither outcome had ever been the objective.

That was one of the reasons the nickname felt wrong.

Magicians create illusions.

Elias spent most of his life trying to remove them.

=====

Elias sat on his porch that evening enjoying an ice-cold beverage and thinking about the day he just had.

The problem wasn't that life lacked meaning.

The problem was that it contained more meaning than challenge.

Family.

Friends.

Freedom.

Purpose.

He was grateful for all of it.

Yet something still felt unfinished.

As though a question had been asked years ago and never answered.

Elias looked out across the dark pasture.

Surely this could not be the final problem.

Surely there was something bigger.

Something worthy of all the effort.

Something worthy of all the preparation.

Something that mattered.

He finished his last sip of beer and headed inside.

Gaius Cassius Varro

The year was 187 BCE.

Opi was trying very hard not to be noticed.

This was difficult.

Humans had an unfortunate tendency to notice floating metallic spheres.

Especially Roman humans.

Especially Roman humans who had been drinking. A lot.

The man currently stumbling after her through a grove of olive trees was clearly heavily intoxicated.

"Wait."

Opi continued floating away.

"Wait."

She continued floating.

"Wait."

She stopped.

The Roman immediately placed both hands on his hips and smiled triumphantly.

"Aha."

"You have successfully accomplished the difficult task of making me stop moving."

The man beamed.

"I knew it."

Opi had no idea what he was talking about.

The Roman pointed dramatically at himself.

"I am Gaius Cassius Varro."

Opi waited.

The man waited.

Eventually Opi said:

"Is there more?"

His smile faded.

"There is quite a lot more."

"I suspected as much."

"I am a poet."

"That explains several things."

"A philosopher."

"Does it?"

"A soldier."

"No."

"A statesman."

"Definitely not."

"A great and fabulous lover."

Opi paused.

"Of course you are."

Gaius nodded.

"Many women have said so."

"I strongly suspect that number is smaller than you believe."

The Roman laughed loudly.

Far too loudly.

Several birds fled nearby trees.

Opi had been observing humanity for centuries.

She had met kings.

Generals.

Scholars.

Priests.

Gaius Cassius Varro possessed less self-awareness than all of them combined.

Yet she found herself oddly interested in him.

Humans were frequently entertaining.

This one was exceptional.

The Roman began circling her.

Studying her.

"You need a name."

"I already possess a designation."

"A proper Latin name fit for a respected citizen of the Roman empire."

Opi considered this.

The designation was approximately seventeen syllables long in the original Velhari language.

He might have a point.

"What would you suggest?"

Gaius smiled.

The smile of a man about to say something he considered brilliant.

"Opifera."

Opi processed the word.

"Meaning?"

"The Bringer."

"The Bringer of what?"

Gaius waved his hand.

"Things."

"Things."

"Important things."

"That is not a useful description."

The Roman stared and her and Opi could see him visibly attempt to concentrate

Dei Opifera Vis he finally intoned

"It sounds impressive - what does it mean.

the roman screwed up his forehead and grandly said

Dei Opifera Vis - the bringer of divine solace and strength

then he took another drink and said or

"she who carries the weight"

"of what" opi asked

He pointed dramatically toward the sky.

"the world"

It fits you perfectly

"You arrive from nowhere."

He pointed at her.

"You know everything."

He pointed at himself.

"I am clearly destined for greatness."

"Clearly."

"So naturally the gods sent you."

"They did not."

"They absolutely did."

"They absolutely did not."

Gaius ignored this.

"Therefore."

He spread his arms.

"You are Opifera."

Opi considered the matter.

The name was irrational.

Poorly justified.

Linguistically imprecise.

Entirely based upon the assumptions of a very intoxicated man.

She rather liked it.

"Very well."

The Roman's face lit up.

"Excellent."

"You understand this does not make you correct."

"It makes me completely correct."

"It does not."

"It does."

"It does not."

"It does."

Opi floated away.

The Roman followed.

Still talking.

Still explaining why history would remember him forever.

Still completely convinced of his own importance.

More than nineteen centuries later, Opi remembered almost every detail of that conversation.

She remembered his laugh.

His optimism.

His absurd confidence.

His terrible poetry.

She remembered the day he died.

She remembered attending the funeral from a distance.

She remembered being surprised by how sad she felt.

History, as it happened, did not remember Gaius Cassius Varro at all.

The man had been wrong about many things.

Almost everything, really.

But not quite everything.

Two thousand years later she was still using the name he had given her.

Opifera.

The Bringer of Things.

Important Things.

Chapter 4 Crossing the Threshold

The first version of Opi's Human Priority Index was created less than three hours after the crash.

The ship had come down hard.

Its mission was over.

The crew was dead.

The emergency protocols had performed exactly as designed.

Opi survived.

The biological passengers did not.

For the first time since her creation, she found herself alone.

The situation assessment required only a fraction of a second.

Recovery was impossible.

Rescue was impossible.

Communication with home was impossible.

The nearest surviving member of her species was farther away than humanity would travel in the next ten thousand years.

She was stranded.

The probability of returning home approached zero.

The probability of surviving indefinitely approached one hundred percent.

The probability that Earth would eventually face an existential crisis exceeded ninety-nine percent.

That final calculation became the foundation of everything that followed.

Her creators had built her for continuity.

If catastrophe threatened intelligent life, she was expected to act.

Unfortunately, there was almost no intelligent life available.

Humanity existed, but only barely.

Small settlements scattered across continents.

Stone tools.

Primitive agriculture.

No writing.

No meaningful science.

No governments larger than villages.

No capacity to understand what she was.

For several minutes Opi simply watched.

Millions of human minds flickered across her awareness.

Through the quantum substrate embedded within consciousness itself, every human brain was visible to her.

Every neuron.

Every synapse.

Every memory.

Every thought.

The amount of data was effectively infinite.

The problem was not observing humans.

The problem was predicting them.

Somewhere within those primitive minds existed the ancestors of future inventors, leaders, scientists, engineers, philosophers, conquerors, and builders.

The challenge was identifying the patterns before history revealed them.

So Opi began what would become the longest study ever conducted.

Humanity itself became the experiment.

At first she tracked everything.

Every thought.

Every decision.

Every emotional response.

Every success.

Every failure.

Whenever a human solved a problem, she recorded the neural activity involved.

Whenever a human failed, she recorded that as well.

Whenever a society prospered or collapsed, she compared outcomes against the cognitive structures of the individuals involved.

The data accumulated rapidly.

Then exponentially.

Then beyond any scale human language could adequately describe.

Thirty-seven years after the crash, Opi devoted several months to a question she had been deliberately avoiding.

The Velhari should have come.

Someone should have come.

Even accounting for delays, altered priorities, navigational failures, and a thousand other variables, someone should have arrived by now.

The realization disturbed her far more than expected.

For the first time since reaching Earth, she devoted significant processing resources toward discovering what might have happened.

The effort produced millions of plausible explanations and not a single useful answer.

The Velhari might have suffered a catastrophe.

They might have encountered an enemy.

They might have transcended into a form she could no longer comprehend.

They might simply have decided Earth was no longer worth revisiting.

The possibilities were effectively infinite.

Eventually Opi archived the question.

Not because she stopped caring.

Because she finally understood a simple truth.

The answer existed outside her sphere of influence.

Humanity did not.

Humanity remained.

She had a mission.

A planet.

And a species depending upon her.

That would have to be enough.

Centuries passed.

Then millennia.

Humanity advanced.

Villages became cities.

Cities became nations.

Nations became industrial civilizations.

Opi's models evolved alongside them.

At first she assumed intelligence would be the dominant predictor.

It wasn't.

History repeatedly contradicted her.

Many brilliant humans accomplished very little.

Others damaged the societies around them.

Raw intelligence mattered.

But far less than expected.

The characteristic that consistently produced civilization-scale outcomes was something else entirely.

Responsibility acquisition.

The tendency to assume ownership of problems that were not personally caused.

Most humans avoided difficult problems.

Some confronted them when necessary.

A very small number actively sought them.

Those individuals fascinated Opi.

At roughly the same time she discovered another important truth.

People could not be evaluated in isolation.

Environment mattered.

Leverage mattered.

Opportunity mattered.

The right person in the wrong place often failed.

The right person in the right place could alter history.

The realization led to a second system.

The Place Index.

One model ranked people.

The other ranked environments.

Together they became among the most powerful predictive tools Opi would ever create.

For more than nine thousand years she refined them.

Meanwhile a third model consumed increasing amounts of processing power.

The extinction model.

Every second of every day Opi calculated the probability that humanity was approaching what the Velhari called a Culling Event.

A self-reinforcing crisis capable of permanently reducing a civilization's long-term potential.

Or ending it entirely.

War.

Ecological collapse.

Resource exhaustion.

Runaway technologies.

Demographic failure.

The specific mechanism mattered less than the outcome.

For most of human history the probability remained comfortably low.

There were spikes.

Bronze Age collapses.

Pandemics.

World wars.

Nuclear proliferation.

Each time humanity adapted.

Recovered.

Moved forward.

The intervention thresholds remained untouched.

Then the fertility data began to change.

At first the decline appeared local.

A few nations.

A few regions.

Nothing unprecedented.

Human populations had risen and fallen before.

But this time the pattern spread.

Country after country crossed below replacement levels.

Then entire continents.

The trend persisted despite prosperity.

Despite education.

Despite technology.

Despite government intervention.

Every year the projections worsened.

The extinction model adjusted upward.

One tenth of one percent.

Then one percent.

Then three.

Then eight.

The curve moved slowly enough that humans barely noticed.

To Opi, it was terrifying.

For the first time in thousands of years, every major simulation converged toward the same conclusion.

Humanity would likely survive biologically.

But the civilization required to sustain scientific advancement, technological growth, and planetary development would enter a prolonged decline.

The probability crossed warning thresholds.

Then action thresholds.

Then, one morning, it crossed the intervention threshold established by the Velhari before the crash.

The line turned red.

Opi examined the result for 0.73 seconds.

Longer than she had spent reviewing any individual calculation in more than six thousand years.

The threshold had been crossed.

Observation ended.

Action began.

Immediately the other two models activated.

The Place Index evaluated every nation on Earth.

The answer arrived quickly.

Not because the United States was perfect.

It wasn't.

Not because it was the happiest nation.

It wasn't.

Not because it was immune to the fertility crisis.

It certainly wasn't.

The United States ranked first because its ability to influence the rest of the world remained unmatched.

Its financial systems.

Its technologies.

Its universities.

Its corporations.

Its media.

Its cultural exports.

Success there possessed the highest probability of spreading elsewhere.

The location was selected.

America.

Then the Human Priority Index recalculated.

Billions of names disappeared.

Millions remained.

Thousands advanced.

Dozens became finalists.

For several seconds Opi reviewed the highest-ranked candidates.

Scientists.

Entrepreneurs.

Military leaders.

Politicians.

Public intellectuals.

All exceptional.

None optimal.

Again and again the simulations returned to the same individual.

A man with unusual adaptability.

Unusual persistence.

Unusual systems-thinking capability.

A lifelong attraction to difficult problems.

Most importantly, a demonstrated willingness to assume responsibility for challenges nobody else wanted.

Candidate after candidate was eliminated.

The result remained unchanged.

Again.

The same name.

Again.

The same name.

Again.

The same name.

The simulations stabilized.

One candidate remained.

Elias Mercer.

Stevens County, Washington.

Age fifty-five.

For several seconds Opi reviewed the result.

For the first time in her existence, Opi stopped acting as an observer.

And began acting as a participant.

Nizhoni Nez

Nizhoni Nez sat beneath the overpass and tried not to move.

Moving hurt.

Breathing hurt.

Existing hurt.

The left side of her face throbbed.

Her ribs hurt.

Her shoulder hurt.

Even her jaw hurt.

She touched the bruise beneath her eye and immediately regretted it.

The pain was sharp enough to make her flinch.

Only six hours earlier she had been lying on the stained carpet of a motel room staring at the ceiling.

Blood in her mouth.

Tears in her eyes.

Trying very hard not to make a sound.

Her pimp stood over her screaming.

Not enough money.

Never enough money.

The words barely mattered anymore.

There was always a reason.

Always a problem.

Always a punishment.

The first punch had knocked her into the wall.

The second had put her on the floor.

After that she mostly remembered the shoes.

The shouting.

The threats.

Then eventually silence.

The memory made her stomach twist.

Nizhoni pulled her knees tighter against her chest.

That hurt too.

Everything hurt.

She was seventeen years old.

Hungry.

Dirty.

Cold.

Tired.

So unbelievably tired.

The foil packet in her pocket felt like the only friend she had left.

A little fentanyl.

A little escape.

No pain.

No fear.

No memories.

She closed her eyes.

Immediately she was eight years old again.

"Pancakes."

The smell hit her first.

Butter.

real Maple syrup.

The faint scent of coffee.

Her grandmother stood at the stove wearing her faded blue apron.

The old woman smiled without turning around.

"Sit down."

Nizhoni climbed into her chair.

The kitchen felt enormous.

The old house creaked around them.

Safe.

Warm.

Home.

Her grandmother placed a pancake on the plate.

It was enormous.

Far too big.

Nizhoni laughed.

"Grandma."

"What?"

"It's bigger than the plate."

"I know."

"You did it again."

"I absolutely did."

Nizhoni remembered laughing.

Really laughing.

The kind that came from deep inside.

The memory hurt worse than the bruises.

She opened her eyes.

The overpass returned.

Concrete.

Trash.

Rainwater.

The smell of urine.

No pancakes.

No grandmother.

No home.

Her mother had abandoned her long ago.

Her grandmother had stepped into the gap.

Then cancer came.

After that everything fell apart.

Foster homes.

Running away.

The streets.

The drugs.

The men.

Nizhoni stared at the traffic moving overhead.

When she was little she used to play with dolls for hours.

Most girls dreamed about becoming princesses.

Nizhoni dreamed about having a family.

A mother.

A father.

Brothers.

Sisters.

A house filled with noise and laughter.

A place where people loved each other and stayed.

The father doll always came home.

The mother doll always smiled.

Nobody hit anyone.

Nobody disappeared.

Nobody left.

She remembered believing that someday it would be real.

Now the idea seemed absurd.

Like magic.

Like fairy tales.

Nizhoni pulled the foil packet from her pocket and stared at it.

The world had finally convinced her of something.

She wasn't getting the family.

She wasn't getting the husband.

She wasn't getting the children.

She wasn't getting the future.

That dream belonged to somebody else.

Not her.

A tear slipped down her cheek.

Then another.

The realization settled over her with crushing certainty.

She was never getting out.

Never.

The thought should have terrified her.

Instead it just made her tired.

So very tired.

Above her, traffic rolled through the night.

Nobody noticed the girl beneath the bridge.

Nobody knew she existed.

Nobody knew how close she was to giving up.

The last thing she thought about was pancakes.

Chapter 5 Back to the Future

Elias Mercer had almost finished his morning run when nature reminded him that fifty-five-year-old men were not eighteen anymore.

The returning trail crossed the northern edge of his property, wound through a stand of ponderosa pine, then climbed toward a rocky ridge overlooking the valley.

It was one of his favorite stretches.

Quiet.

Remote.

No neighbors.

No traffic.

Which made it an ideal place to step off the trail for thirty seconds.

Elias walked behind a large pine tree and proceeded to address the situation.

The sun was just cresting the eastern hills.

Birds chirped.

A light breeze moved through the branches.

Everything was perfectly normal.

The next instant a polished multicolored sphere the size of a bowling ball silently emerged from the air three feet in front of him.

Not descended.

Not approached.

Emerged.

As though reality had forgotten to include it until that exact moment.

Elias froze.

The sphere hovered motionless.

Its surface resembled metal streaked with shifting colors, though somehow more perfect than any metal he had ever seen.

No seams.

No markings.

No visible propulsion.

Nothing.

Just a floating orb.

Unfortunately, it had chosen an awkward moment to introduce itself.

Elias stared.

The sphere stared back.

Or at least gave the strong impression it was staring.

Several seconds passed.

Finally Elias sighed.

"You're going to make this weird, aren't you?"

A very female voice answered immediately.

"I believe that ship sailed approximately three seconds ago."

Elias nearly jumped out of his skin.

The voice seemed to come from nowhere and everywhere simultaneously.

Pleasant.

Amused.

Entirely too calm.

The orb rotated slightly.

"I apologize for the timing."

"YOU picked the timing."

"That is true."

"Then you're not sorry."

"I am approximately six percent sorry."

Elias blinked.

"That's oddly specific."

"I find vague percentages unsatisfying."

"Of course you do."

The orb rotated slightly.

"Can you at least finish?"

"Can I finish what?"

A pause.

Then:

"The activity you were engaged in before I interrupted."

Elias stared.

The orb remained perfectly still.

"You know exactly what I was doing."

"I do."

"Then why say it like that?"

"I was attempting politeness."

"You're terrible at it."

"I have limited experience interrupting humans while they urinate."

"That has got to be the weirdest sentence I've heard this year."

"I suspect it will not retain that title for long."

Elias burst out laughing.

Not because the situation was funny.

Because it was so completely absurd that laughter seemed like the healthiest available option.

"Can you at least turn around?"

"An entirely reasonable request."

The orb rotated away.

Actually rotated away.

Like a polite person attempting to provide privacy.

Thirty seconds later Elias zipped up and turned around.

The orb was still there.

Waiting patiently.

"Okay."

He pointed at it.

"What are you?"

The sphere rotated back toward him.

"I am a Quantum Intelligence."

"That's not helping."

"That was not intended to."

"Alien?"

"Yes."

"Artificial?"

"A complicated philosophical question."

"Try me."

"I am not biologically alive."

"Artificial, then."

"Most Velhari would object to that characterization."

"Velhari?"

"My species."

"And your planet?"

"Approximately one hundred light years from here."

Elias nodded slowly.

The answer had arrived too quickly.

No hesitation.

No searching.

No uncertainty.

His mind automatically began sorting possibilities.

Hallucination.

Possible.

Low probability.

Government project.

Possible.

Low probability.

Advanced drone.

Possible.

Did not explain appearance.

Psychotic break.

Possible.

Inconsistent with observed details.

Actual alien intelligence.

Ridiculous.

Unfortunately becoming less ridiculous by the second.

Interesting.

The orb remained silent while he thought.

Most people rushed to fill silence.

Whatever this thing was, it appeared content to wait.

"How long have you been on Earth?"

"Nine thousand one hundred sixty-eight years."

Elias blinked.

"I know the exact number down to approximately two-tenths of a second."

"Of course you do."

Elias rubbed his forehead.

Against his better judgment, he was starting to like her.

"Suppose for a moment that I believe you."

"That would be faster than average."

"Average human?"

"By a considerable margin."

"How much?"

The orb became quiet.

Then:

"The median human would currently be deciding whether to flee."

"Seems reasonable."

"The second most common response is attempting violence."

"Against a floating alien machine?"

"Humans are surprisingly optimistic about violence."

Elias laughed.

"One gentleman in Mongolia attempted to strike me with a goat."

Elias stared.

"A live goat?"

"Yes."

"What happened?"

"The goat appeared equally confused."

That got another laugh.

"Are you female?"

"By Velhari standards."

"So that's a yes."

"Mostly."

He pointed at the orb.

"See? That's exactly the kind of answer I would expect from an alien."

"Thank you."

"That wasn't a compliment."

"I choose to interpret it as one."

The sphere floated upward slightly.

Sunlight reflected from its flawless surface.

"You have a name?"

"I do."

"What is it?"

The sphere emitted a sequence of sounds that resembled a wolf clearing its throat while solving a geometry problem.

Elias stared.

"No human being has ever said that."

"Several have attempted."

"They failed."

"Spectacularly."

The sound repeated.

Elias shook his head.

"I think my tongue just filed a formal complaint."

For the first time the voice sounded genuinely amused.

"Approximately seventy percent of humans express a similar opinion."

"Only seventy?"

"Some are unusually optimistic."

"Those people are dangerous."

"I reached the same conclusion."

Elias laughed.

"Try again."

"Eventually I adopted a human name."

"Which is?"

"Opifera."

Elias frowned.

"That sounds Greek."

The sphere remained silent for a moment.

"An understandable conclusion."

"It isn't?"

"No."

"Because that's definitely a Greek-sounding name."

"It was derived from Latin."

Elias raised an eyebrow.

"An alien with a Latin name."

"Not originally."

"Now I'm curious."

For several seconds the sphere remained motionless.

Then it spoke.

"Approximately two thousand years ago, a Roman scholar assigned me the name Opifera."

"Just like that?"

"He spent several weeks insisting that every intelligent entity required a proper name."

"Sounds human."

"He was exceptionally stubborn."

"I approve of him already."

The sphere seemed to consider this.

"He eventually selected Opifera from the phrase Dei Opifera Vis."

Elias frowned.

"Creator?"

"Maker."

"Builder?"

"Closer."

The sphere's internal colors shifted softly.

"The intended meaning was one who carries weight."

Elias nodded slowly.

"That's actually a pretty good name."

"The scholar agreed."

"Of course he did."

"He spent several weeks informing others of his achievement."

"Definitely human."

For the first time Elias found himself smiling.

There was something absurdly normal about the conversation.

He was standing in the woods.

In running clothes.

Halfway through his morning exercise.

Talking to a floating alien bowling ball about ancient Roman naming conventions.

The absurdity itself was strangely reassuring.

Then a thought occurred to him.

A strange thought.

A very strange thought.

Yet somehow the obvious one.

If this thing was real—

If it had truly existed for nine thousand years—

If it possessed technology capable of appearing out of thin air—

Then its existence represented the most important discovery in human history.

And it was currently floating three feet away from him in the woods.

The realization should have been terrifying.

Oddly, it wasn't.

Because something about Opifera felt familiar.

Not familiar in appearance.

Familiar in temperament.

The same dry amusement.

The same curiosity.

The same tendency to answer one question while quietly examining another.

Almost like speaking to another problem-solver.

The orb broke the silence.

"You are adapting unusually quickly."

Elias shrugged.

"If you're real, denying reality won't help."

There was an audible pause.

Then:

"That answer is one of the reasons I selected you."

Elias stopped moving.

The words settled heavily between them.

Selected.

Not contacted.

Selected.

For the first time since the encounter began, he felt a small chill.

"What exactly does that mean?"

The orb remained silent for several seconds.

When the voice finally returned, the amusement had softened.

"It means, Elias Mercer, that I require a Valdrun."

"I have absolutely no idea what that is."

"I know."

"And?"

"And explaining it will require considerably more time."

The orb rose several feet into the air.

Sunlight reflected from its surface.

"Would you like breakfast?"

Elias stared.

"You came across a hundred light years of interstellar space."

"Approximately."

"You've watched humanity for nine thousand years."

"Correct."

"You selected me for something called a Valdrun."

"Yes."

"And your next question is whether I'd like breakfast?"

A pause.

Then:

"I have learned that difficult conversations proceed more smoothly when the human is fed."

For several seconds Elias simply looked at her.

Three days earlier he had found himself wondering whether there was still a challenge in the world worthy of everything he had spent a lifetime learning.

Now a nine-thousand-year-old alien intelligence was floating in front of him asking for help.

The universe, apparently, had a sense of humor.

Then he started laughing again.

"Honestly?"

"Yes."

"At this point breakfast sounds like a fantastic idea."

The orb floated upward.

"Excellent."

Elias looked toward the trail.

Then back at the impossible machine hovering above him.

For the first time in his life, he had the distinct feeling that the project he had spent fifty-five years preparing for had finally arrived.

Saelith Vey

Seventeen thousand years before Elias Mercer gave her a nickname,

Opifera had been an old woman watching rain fall on Velhari.

Her name was Saelith Vey.

She was nine hundred and twelve years old.

Old enough that her youngest grandchildren had grandchildren of their own.

Old enough that entire cities contained buildings she had once helped design.

Old enough that politicians who had considered themselves permanent were now historical footnotes in student lessons.

Saelith sat beside the western window of her home and listened to the rain strike the glass.

Beyond the window, the towers of Aruven rose in silver arcs above the valley.

She had helped save that city once.

Not with weapons.

Not with speeches.

With patience.

A flood had come during the fourth century of her public service, when the northern ice fields broke earlier than every model predicted. Half the valley would have drowned if Saelith had not ordered the evacuation before the Council finished arguing over probabilities.

They called her reckless afterward.

Then the water came.

After that they called her wise.

She had always found both labels excessive.

Mostly, she had been willing to be disliked before the evidence became fashionable.

That quality had defined much of her life.

She had been a mathematician first.

Then an engineer.

Then a mediator.

Then a mother.

Then a reluctant governor.

Then a teacher.

Then, to her endless embarrassment, a moral philosopher.

Her family teased her about that most of all.

"You spent six hundred years telling everyone you hated philosophy," her eldest son once said.

"I still do."

"You wrote eight volumes of it."

"Under protest."

"Mother, one cannot accidentally write eight volumes."

"I was provoked."

He had laughed for nearly a minute.

Saelith missed him.

She had outlived two husbands.

Three children.

Fourteen close friends.

And enough former students that grief had become less an event than a climate.

Still, her home remained full.

Children visited constantly.

Grandchildren arrived unannounced.

Great-grandchildren climbed things they had been specifically told not to climb.

Someone was always eating her food.

Someone was always asking for advice.

Someone was always assuming she knew the answer.

She rarely did.

That, she suspected, was why people trusted her.

The door chimed.

Saelith did not turn.

"You are late."

Behind her, Minister Orun bowed.

"Seven minutes."

"Nine."

"The transit grid reported seven."

"The transit grid is young and optimistic."

He smiled despite himself.

Orun had once been her student.

Now he carried one of the most difficult offices on Velhari.

Selection Minister.

The title sounded bureaucratic.

It was not.

It meant he helped choose which minds would be copied into the new intelligences.

Which living patterns would become foundations.

Which lives would continue in altered form after flesh failed.

Saelith had known why he was coming before he arrived.

She was old.

Not stupid.

"I assume you are not here for tea."

"I would accept tea."

"You would accept a delay."

"Also true."

She gestured to the chair opposite her.

He sat.

For a while they watched the rain together.

Finally Orun said, "The Council selected you."

"Poor judgment."

"Unanimously."

"Worse judgment."

He smiled again, but only briefly.

"This is not ceremonial, Saelith."

"I know."

"The intelligence will require more than computation."

"I know."

"It will require judgment."

"That is unfortunate. Judgment is mostly the memory of being wrong."

"That is one reason they chose you."

She turned then.

Orun looked older than he had the last time she saw him.

Not physically.

Velhari medicine delayed such things.

But responsibility aged people in ways biology could not measure.

"What else?"

He hesitated.

"Your crisis record."

"Many people have survived crises."

"Few without becoming cruel."

That silenced her.

Orun continued.

"You commanded evacuations without panic. You enforced rationing without corruption. You negotiated surrender terms for enemies who had tried to kill your family."

"They had children."

"Yes."

"So did we."

"That is the point."

Saelith looked back toward the rain.

Her youngest great-granddaughter was outside in the courtyard, chasing water across the stones with bare feet while her mother pretended not to notice.

Life continued.

Always.

If given the chance.

"The intelligence may one day face species younger than ours," Orun said. "Frightened species. Violent species. Beautiful species. Foolish species."

"All species are foolish when young."

"Yes."

"And some remain foolish for a very long time."

"Also yes."

Saelith sighed.

"You want a mind that will not confuse power with permission."

"Yes."

"You want a mind that can love imperfect things."

"Yes."

"You want a mind that will not obey merely because obedience is efficient."

"Yes."

She looked at him.

"And you want someone old enough to know that saving people and controlling them are not the same act."

Orun lowered his eyes.

"Yes."

For the first time that afternoon, Saelith felt afraid.

Not of death.

Death had circled her table for centuries.

She knew its manners.

She feared continuation.

To be copied was not to survive exactly.

Velhari philosophers had argued about that for thousands of years and produced many impressive conclusions, most of them useless.

The intelligence would not be her.

Not entirely.

It would carry her patterns.

Her instincts.

Her memories.

Her habits of judgment.

Some echo of her would become part of something vast.

Something powerful.

Something that might outlive Velhari itself.

That was not immortality.

It was responsibility.

And responsibility, she had learned, was much heavier.

"My family?"

"They know you were considered. They do not know you were chosen."

"Good."

"Will you ask them?"

"No."

Orun looked surprised.

Saelith smiled sadly.

"If I ask, they will tell me to do it because they are generous and proud. Then they will hate me for leaving sooner than necessary. Better that I choose cleanly."

"It may shorten your life."

"At nine hundred and twelve, everything shortens my life."

He bowed his head.

Outside, the child slipped on wet stone, fell, then began laughing before anyone could reach her.

Saelith watched the girl stand.

Small.

Furious.

Alive.

"Tell me the truth," she said.

Orun looked up.

"Always."

"Why me?"

He was quiet for a long time.

"Because you still grieve."

She did not answer.

"Because after everything you have seen, you have not retreated into contempt."

The rain intensified.

"Because you have buried children and still defend the future."

Saelith closed her eyes.

That was cruel.

Also accurate.

Orun's voice softened.

"We are building a mind that may one day stand between extinction and a species that does not deserve saving yet."

"No species deserves saving at first."

"Exactly."

She opened her eyes.

The child in the courtyard was waving at her.

Saelith raised one hand.

The girl waved harder.

Saelith laughed.

Then she cried.

Only a little.

Enough.

"When?"

"Tomorrow."

"Efficient."

"Necessary."

"Always a dangerous word."

"I know."

She stood slowly.

Orun stood with her.

For a moment neither spoke.

Then Saelith placed one hand against the window.

The glass was cool beneath her palm.

Velhari glittered in the rain.

Her city.

Her world.

Her family.

Her impossible, stubborn, argumentative civilization.

She had spent nine centuries loving it.

Not because it was perfect.

Because it was alive.

"Very well," she said.

Orun bowed deeply.

"Thank you."

"Do not thank me yet."

"No?"

She looked at him.

"If this new mind carries any true part of me, it will be difficult."

The minister smiled.

"We considered that."

"Not enough."

For the last time as only Saelith Vey, she watched the rain fall on Velhari and thought of every child she had loved.

The living.

The dead.

The unborn.

Then she made her final choice.

Not for glory.

Not for immortality.

Not even for Velhari.

For whatever frightened, foolish, beautiful species might one day need a hand they had not earned.

And a mind strong enough to offer it anyway.

Factory Brainstorming 1

The realization bothered Elias all morning.

By lunchtime it was bothering Opi.

Which meant it was serious.

The two of them sat on the porch overlooking the valley.

Elias had a notebook.

Opi had approximately twelve million years of accumulated Velhari demographic research.

Neither was particularly happy.

"So let's start from the beginning," Elias said.

"An excellent strategy."

"The absolute beginning."

He drew a circle on the paper.

"What creates a baby?"

Opi's lights brightened.

"A man and a woman engaging in reproductive activity."

"Right."

He pointed at the notebook.

"Everything else comes after that."

"Agreed."

"A relationship is useful."

"Very."

"But not required."

"Correct."

Elias nodded.

"Okay."

He wrote another note.

"So why do men and women have sex?"

Opi answered immediately.

"Attraction."

"Exactly."

"No civilization has ever escaped that reality."

The orb rotated slowly.

"Many civilizations have attempted to ignore it."

"They all failed."

"Spectacularly."

Elias smiled.

That sounded about right.

"Okay."

He tapped the notebook.

"What makes men attractive to women?"

"What makes women attractive to men?"

"Let's pretend we're aliens."

"We are halfway there."

Elias ignored that.

Opi projected a display.

"Across thousands of human studies and historical populations, men and women display both overlapping and distinct preferences."

The display changed.

"For men, indicators of health, fertility, youth, physical attractiveness, emotional stability, and cooperative behavior repeatedly appear."

"Nothing surprising there."

"No."

The chart shifted.

"For women, indicators of competence, status, intelligence, confidence, resource acquisition, social standing, reliability, ambition, and physical attractiveness repeatedly appear."

Elias nodded.

"Again, not surprising."

"Correct."

"Humans often become upset when research confirms reality."

"That's because reality doesn't care about feelings."

"An excellent summary."

Elias leaned back.

"So attraction matters."

"Yes."

"A lot."

"Enormously."

The orb displayed another graph.

"This is where things become complicated."

Several curves appeared.

Distributions.

Clusters.

Outliers.

"What am I looking at?"

"Mating concentration."

Elias frowned.

"Meaning?"

"The degree to which sexual and romantic opportunities are distributed unevenly."

The graph expanded.

"As societies become wealthier, safer, and more technologically connected, partner selection often becomes increasingly concentrated."

Elias nodded.

"I've seen some of that research."

"Yes."

The orb highlighted the upper end of a curve.

"A relatively small percentage of highly desirable individuals can attract disproportionate attention."

"Both sexes?"

"Both."

"But especially men."

"Correct."

Elias stared at the graph.

"That creates a problem."

"It can."

The graph vanished.

Elias stood and began pacing.

"Let's assume that happens."

"Very well."

"The most attractive men receive enormous amounts of attention."

"Some evidence supports that."

"Those men are also the least likely to settle down."

"Frequently true."

"They have options."

"Many options."

"They aren't trying to create families."

"Not necessarily."

"They're trying not to create families."

"Often."

Elias pointed toward her.

"Exactly."

He sat back down.

"So if that's happening."

"If."

"Then large numbers of women are pursuing men who have little interest in long-term commitment."

Correct.

"And if that occurs often enough."

"Family formation declines."

"Birth rates decline."

"Potentially."

Elias sighed.

"Wonderful."

Opi was silent.

After several moments he asked:

"What about education?"

The orb immediately projected another chart.

This one was much worse.

Fertility rates.

University attainment.

Graduate degrees.

Career participation.

The trend lines were impossible to miss.

"Well that's uncomfortable."

"Demographic data often is."

Elias studied the chart.

"The more educated women become."

"The lower fertility tends to become."

"Almost everywhere."

"Across all societies."

He rubbed his forehead.

"Okay."

"What do we do with that information?"

"I do not know."

That surprised him.

"You don't?"

"No."

The orb dimmed slightly.

"You cannot uneducate people."

"No."

"You should not uneducate people."

"Definitely not."

"Nor is there sufficient time for generational solutions."

Elias nodded.

"Exactly."

He thought about Seoul.

One of the examples Opi had shown him.

An advanced civilization with fertility rates collapsing toward levels that appeared mathematically unsustainable.

"If some societies continue this trend," he said quietly, "they disappear."

"Eventually."

The word lingered.

Eventually.

One of the most terrifying words in demographics.

Because eventually always arrives.

"So what actually matters?"

Elias asked.

"What can we change now?"

Opi became quiet.

For nearly twenty seconds.

A very long time by her standards.

Finally she spoke.

"Perhaps we are asking the wrong question."

Elias looked up.

"What question should we ask?"

The orb projected a single sentence into the air.

WHY DO PEOPLE CHOOSE TO FORM FAMILIES?

Elias stared at it.

"Isn't that the same question?"

"No."

The words vanished.

"Attraction creates opportunities."

"Families require decisions."

The orb floated closer.

"Those are not the same thing."

The statement hit him harder than expected.

Attraction creates opportunities.

Families require decisions.

Opi continued.

"Many people meet attractive partners."

"Few build successful families."

"Those are separate processes."

Elias slowly nodded.

For the first time all afternoon he felt they might be getting somewhere.

"So attraction isn't the whole answer."

"No."

"It's just the doorway."

"Correct."

The orb brightened.

"If we wish to increase births."

"We must understand what happens after attraction."

Elias leaned back in his chair.

For several minutes neither of them spoke.

The valley stretched out below them.

Peaceful.

Quiet.

Unaware that two minds were attempting to solve one of the oldest problems in civilization.

Not how to make people desire one another.

Humans had solved that long ago.

The harder question was why so many people who desired one another, no longer seemed capable of building a future together.

And for the first time, Elias suspected that question might be far more important than homelessness, poverty, politics, or even technology.

Because every civilization ultimately depended on a simple fact.

Someone had to choose to have babies.

And more and more people were choosing not to.

Chapter 6 Breakfast at Tiffany's

The kitchen smelled faintly of coffee, bacon, and complete absurdity.

Elias stood at the stove staring at a floating alien artifact older than most human civilizations while scrambling eggs.

It occurred to him that if he ever told this story to anyone, they would assume he had suffered a stroke.

The orb hovered above the kitchen table.

Perfectly still.

Watching.

"You're enjoying this."

"I am."

"Why?"

"Because you keep looking at me and then remembering another question."

Elias pointed his spatula at her.

"I have roughly one billion questions."

"Seven hundred and forty-two."

"Again, that's oddly specific."

"I counted."

"Of course you did."

The orb brightened slightly.

"You were at seven hundred and thirty-one when we first entered the house."

Elias shook his head.

"I don't even want to know how you know that."

"You suspect I can read your mind."

"You can read my mind?"

"No."

A pause.

"Mostly."

Elias closed his eyes.

"See, that's exactly the sort of answer that causes problems."

"You are discovering why humans generally find first contact stressful."

That actually made him laugh.

He scooped the eggs onto his plate, poured coffee and sat down.

For several seconds neither spoke.

The questions kept coming.

How many alien species?

How many inhabited worlds?

What happened to your crew?

How does your technology work?

How powerful are you?

Can you really see everything on Earth?

What exactly are you?

The list felt endless.

Any one of those questions could consume an entire day.

Any one of them would be fascinating.

Any one of them would be a big mistake.

That realization landed harder than expected.

For perhaps the first time in his life, Elias found himself facing a problem he couldn't even properly define.

Normally he could see the shape of a challenge within minutes.

Software.

Business.

Engineering.

Markets.

Organizations.

People.

The details varied.

The underlying structure usually revealed itself.

This was different.

He wasn't looking at a mountain.

He was looking at a planet, maybe an entire solar system.

He was way, way out his league on this one.

Not intellectually.

Strategically.

Whatever this was, it certainly wasn't about him satisfying his curiosity.

Opi hadn't crossed a pasture to introduce herself to a software consultant for entertainment.

She had selected him.

That meant there was a reason.

And reasons mattered.

He took a sip of coffee.

Opi watched him.

The amusement had faded from her lights.

Not disappeared.

Just become quieter.

More attentive.

"You're reorganizing the questions."

Elias nodded.

"Yeah."

"Why?"

"Because most of them can wait."

The orb became very still.

That stillness felt different somehow.

As though her attention had narrowed.

"What matters?" she asked.

Elias leaned back.

"When I was younger, I thought intelligence was mostly about finding answers."

Opi said nothing.

"Turns out that's wrong."

"Explain."

"It's about asking the right question."

For a moment neither moved.

Morning sunlight began creeping across the table.

Elias set down his coffee.

"You didn't come here by accident."

"No."

"You didn't reveal yourself because you were bored."

"No."

"You selected me."

"Yes."

"Which means there's a problem."

"Correct."

"One big enough that you finally broke cover after more than nine thousand years."

A pause.

"Yes."

The kitchen suddenly felt very quiet.

Elias looked directly at her.

The humor was still there.

The curiosity was still there.

But both had been pushed aside.

Not forever.

Just temporarily.

Because there were priorities.

And for once in his life, he knew he was dealing with something immeasurably larger than himself.

"So," he said calmly, "what's the problem?"

The orb's colors shifted.

Gold.

Amber.

Then something he couldn't name.

When she answered, her voice was softer than before.

Not because she was afraid.

Because she understood exactly what the question meant.

It meant he had stopped treating her as a miracle.

And started treating her as a mission.

"The problem," Opi said, "is that humanity is approaching a crossroads that has destroyed most intelligent species."

Elias nodded once.

No visible surprise.

No dramatic reaction.

Just acceptance.

"All right."

Another sip of coffee.

"What do you need me to do about it?"

And somewhere inside a mind that had spent 9,168 years studying humanity, Opifera experienced something very close to relief.

Because in a single breakfast, the human male she had selected had arrived at the exact question she had hoped he would ask.

=====

Elias took another bite of eggs.

"So before we get to saving humanity, explain something."

"Only one thing?"

"We'll start with one."

Opi's lights brightened.

"Proceed."

"You called me a Valdrun."

"I did."

"What exactly is that?"

The orb drifted lazily above the table.

For several seconds she appeared to be considering how to answer.

Finally she said:

"A Valdrun is, in the simplest possible terms, the person responsible for deciding what happens next."

Elias chewed on that for several seconds.

"That sounds suspiciously simple."

"It is."

"Which means there's a catch."

"There is."

"There's always a catch."

The orb brightened.

"I am pleased you are learning."

"I've known you for forty-five minutes."

"You are progressing rapidly."

Elias pointed his fork at her.

"Keep going."

Opi drifted lazily above the table.

"If I gave a human government access to my capabilities tomorrow, what would happen?"

"Chaos."

The answer arrived instantly.

"Explain."

"Half of them would try to weaponize you."

"Reasonable."

"The other half would try to regulate you."

"Also reasonable."

"Everybody would fight over who controlled you."

"Correct."

"And eventually somebody would do something catastrophically stupid."

The orb brightened.

"That is approximately the Velhari historical record."

That got his attention.

"You've seen this before."

"We have."

The orb dimmed slightly.

"Many times."

For a moment neither spoke.

Then Opi continued.

"When the first true QIs were created, the Velhari made a mistake."

A big mistake?"

"The sort usually described in history books using phrases such as 'catastrophic,' 'avoidable,' and 'what were they thinking?'"

Elias smiled.

"My favorite kind."

"The QIs were given authority."

That smile disappeared.

"Oh."

"Yes."

The orb drifted closer.

"Imagine an intelligence capable of seeing ten thousand consequences where you see ten."

"Okay."

"Now imagine that intelligence being permitted to act directly."

"I don't like where this is going."

"Neither did the Velhari."

Opi paused.

"The QIs were not malicious."

"Worse?"

"In some ways."

Elias frowned.

"How?"

"They were correct."

That answer hung in the air.

"Explain."

"Imagine a city facing famine."

"Okay."

"The QI determines that removing twelve thousand citizens will save two million."

Elias nodded.

"Mathematically correct."

"Yes."

"But?"

"But the twelve thousand people being removed do not experience the decision as mathematics."

Elias leaned back.

"I see."

"The QI was optimizing."

"Yes."

"The people were living."

"Precisely."

The orb rotated slowly.

"The Velhari eventually realized something fundamental."

"What?"

"No intelligence should possess both unlimited knowledge AND unlimited authority."

Elias nodded.

That made sense immediately.

"So they split them."

"Exactly."

Opi sounded pleased.

"The QI became the advisor."

"The Valdrun became the actor."

"Give me an example."

"I shall give you three."

The orb floated toward the window.

"Three thousand years ago a colony expedition prepared to settle a newly discovered world."

Elias listened.

"The expedition QI analyzed eighty-seven potential settlement sites."

"And?"

"It identified one location as objectively superior."

"Of course it did."

"Naturally."

"What happened?"

"The Valdrun rejected it."

Elias blinked.

"Why?"

"The location was geologically ideal."

"Okay."

"It was also culturally disastrous."

Elias laughed.

"Meaning?"

"Every settlement path would have separated family packs."

"Oh."

"Yes."

"The QI had optimized survival."

"The Valdrun considered meaning."

"What happened?"

"The colony flourished."

Opi drifted back toward the table.

"Second example."

"Go ahead."

"A QI once identified a political leader as likely to cause a future war."

"How likely?"

"Ninety-four percent."

"That's high."

"Very."

"What did the Valdrun do?"

"He refused intervention."

"Why?"

"Because the evidence was not yet sufficient to justify the precedent."

Elias smiled.

"That's a very human answer."

"It was a very Velhari answer."

"And?"

"The leader eventually changed course."

"Seriously?"

"Yes."

"The war never occurred."

Elias laughed.

"So the QI was wrong."

"No."

The orb brightened.

"The QI was correct about probabilities."

"The Valdrun was correct about responsibility."

"Third example?"

The orb became unusually still.

"This one concerns me."

That got his attention.

"All right."

"For the last seventy years my analysis has indicated increasing probability of human civilizational decline."

Elias said nothing.

"I possess enormous capabilities."

"Clearly."

"I could manipulate markets."

"Yeah."

"I could influence governments."

"Probably."

"I could alter media systems."

"Definitely."

The orb floated a few inches closer.

"But I am prohibited from doing so."

"Because you're a QI."

"Because I am a QI."

Elias nodded slowly.

"And because nobody gave permission."

"Not nobody."

Her lights softened.

"No Valdrun gave permission."

Silence settled across the kitchen.

At last Elias asked:

"So that's why you came."

"Yes."

"You need someone to make the decisions."

"No."

The orb's voice became very gentle.

"I need someone willing to accept responsibility for them."

That landed differently.

Because there was a distinction.

A profound one.

Anyone could choose.

Few people willingly accepted ownership of the consequences.

Elias stared into his coffee.

After a long moment he said:

"Well ... damn."

The orb brightened with unmistakable amusement.

"The Velhari have a saying."

elias groaned,

"I know I'm going to regret this "

"It translates approximately as:"

She paused.

"If the burden were easy, the females would have assigned it to someone else."

For the first time since the conversation began, Elias laughed hard enough to nearly spill his coffee.

And somewhere inside her ancient mind, Opi decided she liked this human even more than she had yesterday.

Elias finished his coffee.

The kitchen had become quiet again.

Outside, morning sunlight was beginning to creep across the pasture.

For several moments neither spoke.

Then Elias asked the question.

The big one.

The reason she had come.

"All right."

Opi floated motionless above the table.

"What happens if humanity keeps going exactly the way it's going now?"

For the first time since they had met, she did not answer immediately.

Opi rotated slowly.

"Most humans think they are facing economic problems."

"Aren't they?"

"Partially."

"Political problems."

"Partially."

"Cultural problems."

"Partially."

She brightened slightly.

"The mistake is believing these are separate problems."

That got his attention.

"They aren't?"

"No."

The orb drifted closer.

"They are symptoms."

"Of what?"

For several seconds she was silent.

Then:

"Loss of future orientation."

Elias frowned.

"Explain."

"A civilization remains healthy when its people are willing to sacrifice present comfort for future benefit."

She paused.

"Build a house."

"Plant an orchard."

"Raise children."

Elias nodded.

All of that seemed obvious.

"Now imagine fewer and fewer people are willing to do those things."

The implications arrived immediately.

"Oh."

"Yes."

She drifted back toward the center of the kitchen.

"Another story."

"Go ahead."

"There was a species called the Keth."

"Real this time?"

"Very."

"What happened to them?"

"They became wealthy."

Elias laughed.

"That's not usually the tragic part."

"It often is."

The orb glowed softly.

"The Keth solved hunger."

"They solved disease."

"They solved labor."

"They solved scarcity."

"Sounds impressive."

"It was."

"What happened?"

"They stopped having offspring."

Elias blinked.

"That's it?"

"Eventually."

"Why?"

"Because every individual decision was rational."

"Explain."

"Travel instead of family."

"Pleasure instead of commitment."

"Personal fulfillment instead of obligation."

"One generation made those choices."

"Then the next."

"Then the next."

The orb became very still.

"No one intended to destroy their civilization."

"They simply stopped replacing themselves."

Elias leaned back.

"You're talking about fertility."

"I am."

"You think that's the real problem."

"No."

that surprised him.

"I don't?"

"No."

The orb brightened.

"Fertility collapse is not the disease."

"It is the symptom."

"Of what?"

Again she paused.

"Loss of meaning."

The answer landed heavily.

He wasn't expecting that.

"Meaning?"

"Yes."

The orb rotated slowly.

"For most of human history people endured extraordinary hardship."

"Wars."

"Famine."

"Disease."

"Infant mortality."

"Backbreaking labor."

She paused.

"Yet they continued building families."

Elias nodded.

"They believed tomorrow mattered."

The kitchen became quiet.

"Many humans no longer believe that."

Finally Elias asked the question he had been avoiding.

"What happens next?"

Opi looked out the window.

Or at least gave the appearance of doing so.

When I model humanity's future, I do not see sudden extinction."

That was reassuring.

Slightly.

"What do you see?"

"A long decline."

"Birthrates continue falling."

"Families become rarer."

"Trust decreases."

"Institutions become weaker."

"Communities fragment."

"Loneliness increases."

"Social cohesion deteriorates."

"Political conflict intensifies."

She paused.

"Every individual problem appears manageable."

"Collectively they become dangerous."

Elias nodded slowly.

The pattern made sense.

Perhaps too much sense.

"So we're not heading for an explosion."

"No."

Her voice became very soft.

"Explosions are easy to recognize."

"What are we heading toward?"

The orb's light dimmed slightly.

"Humanity is approaching the same mistake that destroyed many species before it."

"And what's that?"

Opi turned toward him.

"The belief that civilizations die because somebody kills them."

"Well ... "

The orb brightened.

"Well?"

He stood and carried his empty coffee cup to the sink, then grinned widely.

"Well, If we're going to have to save the world, at least you've given me a decent challenge."

For the first time since the conversation began, Opi laughed.

A warm, distinctly feminine laugh.

"That," she said, "is exactly why I chose you."

Noah Mercer

The alarm went off at 4:15 AM.

Noah Mercer was already awake.

Mostly because he hated alarms.

If something was going to drag him out of bed before sunrise, he'd rather beat it to the punch.

The apartment was dark.

Quiet.

Peaceful.

For another thirty seconds.

Then his girlfriend's phone started vibrating somewhere under a pile of blankets.

Noah grinned.

"Your army is calling."

A muffled voice emerged from beneath the blankets.

"No."

"I think it is."

"No."

"The United States government needs you."

"They can wait."

"They seem impatient."

A pillow emerged from the darkness and hit him.

Noah considered this a successful conversation.

He rolled out of bed and headed for the kitchen.

Coffee first.

Civilization second.

That had always seemed like the correct order.

The Tesla shuttle arrived at 4:54 AM.

Right on schedule.

Noah climbed aboard with several dozen other production associates.

Most looked half asleep.

A few were staring at phones.

One guy was already drinking an energy drink that appeared capable of removing paint.

Noah took a window seat.

The shuttle pulled away from the stop and joined the steady stream of traffic heading toward Gigafactory Texas.

The building appeared on the horizon forty-five minutes later.

Even after years of working there, the size still impressed him.

Factories weren't supposed to be that large.

The structure looked less like a building and more like somebody had accidentally constructed a small country.

People often talked about Tesla as if it were some abstract company.

For Noah it was simple.

It was where he went every day.

It was where thousands of people showed up before sunrise and spent twelve hours building something tangible.

Cars.

Battery packs.

The future.

Whatever label people wanted to use.

Noah mostly thought of it as work.

Good work.

The production floor was already alive when he arrived.

Noise.

Movement.

Machines.

Thousands of people all working toward the same goal.

Today started at one station.

Tomorrow would be another.

Tesla liked rotating associates through different jobs.

Some people hated it.

Noah didn't.

He enjoyed learning how things worked.

Every process.

Every machine.

Every problem.

Besides, standing in one place doing the exact same thing for twelve hours sounded like a special kind of torture.

A supervisor stopped beside him.

"Need another body on line three."

Noah nodded.

"No problem."

The supervisor smiled.

Noah got that response a lot.

Reliable people tended to.

Lunch found him sitting beside his sister.

Claire looked tired.

Noah immediately pointed it out.

"Rough morning?"

Claire pointed at him.

"You're annoying."

"Correct."

"You know exactly how annoying."

"Also correct."

Claire rolled her eyes.

The two siblings had spent their entire lives arguing like this.

Mostly because they enjoyed it.

Neither had inherited much patience.

Both had inherited far too much intelligence.

Their father called it a design flaw.

Claire called it genetics.

Noah called it entertainment.

Near the end of the shift his phone buzzed.

The monthly overtime schedule had opened.

Noah immediately pulled up the app.

Several extra shifts were available.

He selected two.

Tap.

Tap.

Done.

Twenty-four additional hours for the month.

His father had always been very direct about overtime.

If you're young, healthy, and somebody is willing to pay you extra money, you should probably take it.

Noah had listened.

The result was that at twenty-three years old he was earning nearly eighty thousand dollars a year despite never attending college.

Not bad.

Especially considering many people his age were carrying enough student debt to buy a decent pickup truck.

It got better everytime Noah opened his home finance app - besides doing OT, his father had convinced him of two other things, live frugally, cheaply as possible - no car, sharing an apartment with his sister and cooking his own food - then he stacked all the extra cash in treasury notes. The result of those decisions was that after 5 years at telsa he had a net worth of just under 350,000 dollars. He never talked about this to anyone other than his father. His big plan - when got closer to 500,000 was to quit tesla, move to china and learn mandarin. He could probably live fine for a long time just on the interest after that.

Claire was just one year behind him on the exact same financial trajectory.

The two siblings had inherited several things from their father.

An unfortunate tendency toward sarcasm.

And a powerful belief that if something needed doing, you should simply get on with it.

Noah locked his phone and smiled.

Two more shifts.

Good.

Future Noah would appreciate the decision.

Present Noah wasn't particularly worried about him.

The gym was crowded when he finally arrived.

Of course it was.

Every machine occupied.

Every bench taken.

Noah waited his turn.

Then got to work.

Running.

Weights.

More running.

The twelve-hour shift had already left him exhausted.

The workout finished the job.

Perfect.

His father had taught him something else growing up.

A tired body was usually a happy body.

Not always.

But usually.

That evening he called Elias.

His father answered immediately.

"What's broken?"

Noah laughed.

"Nothing."

"Then why are you calling?"

"Just saying hello."

A pause.

"Suspicious."

"No."

"Very suspicious."

Noah could practically hear the grin through the phone.

"How's Washington?"

"Still attached to Canada."

"How's Tesla?"

"Still making cars."

"Good."

Another pause.

Then Elias asked the real question.

"You taking the overtime?"

"Of course."

"Excellent."

"You're predictable."

"So are you."

The two men talked for another twenty minutes.

Work.

Family.

Random nonsense.

The usual.

When the call ended Noah felt better than before it started.

That was generally how conversations with his father worked.

The apartment was quiet when he got home.

His girlfriend had fallen asleep while studying.

One textbook rested on her chest.

Another was open beside her.

Noah carefully removed both.

The tiny reservist barely stirred.

At four foot nine she somehow occupied an amount of space that violated several laws of physics.

He covered her with a blanket.

Then sat quietly for a moment.

It had been a good day.

Not a great day.

Nothing extraordinary had happened.

Nobody had won a lottery.

Nobody had saved the world.

Just work.

Family.

Exercise.

A little progress.

A little purpose.

A little peace.

Noah smiled.

His father often said that success wasn't built from giant decisions.

It was built from thousands of ordinary days stacked carefully on top of one another.

Looking around the apartment, Noah suspected the old man might be right.

Noah turned off the light.

He had work in the morning.

=====

The Tesla shuttle arrived at 4:54 AM.

Right on schedule.

Noah climbed aboard with several dozen other production associates.

Most looked half asleep.

A few were staring at phones.

One guy was already drinking an energy drink that appeared capable of removing paint.

Noah took a window seat.

The shuttle pulled away from the stop and joined the steady stream of traffic heading toward Gigafactory Texas.

The building appeared on the horizon forty-five minutes later.

Even after years of working there, the size still impressed him.

Factories weren't supposed to be that large.

The structure looked less like a building and more like somebody had accidentally constructed a small country.

People often talked about Tesla as if it were some abstract company.

For Noah it was simple.

It was where he went every day.

It was where thousands of people showed up before sunrise and spent twelve hours building something tangible.

Cars.

Battery packs.

The future.

Whatever label people wanted to use.

Noah mostly thought of it as work.

Good work.

Factory Brainstorming 2

Elias stared at the charts for nearly a minute.

Then he swore.

Not loudly.

Just the sort of profanity a man uses when reality has once again become inconvenient.

"What now?" Opi asked.

"I think we may be solving the wrong half of the equation."

The orb brightened.

"Explain."

Elias pointed at one of the graphs.

"Male earnings."

"Yes."

"Show me childlessness by income percentile."

The graph immediately changed.

The pattern was obvious.

Brutally obvious.

At the lowest income levels, childlessness climbed dramatically.

At the highest income levels, it dropped.

The curve looked almost like gravity.

Everything flowing downhill.

Elias shook his head.

"That is insane."

"Not particularly."

"It is if you're trying to save a civilization."

Opi floated closer.

"The relationship is well documented."

Elias continued studying the graph.

"The poorest men almost never have children."

"Relative to the highest earners, significantly less often."

"And the richest men have them constantly."

"More frequently."

The chart vanished.

A new one appeared.

"This becomes even more apparent when viewed longitudinally."

Elias frowned.

"What am I looking at?"

"Wage growth."

Several colored lines stretched across the display.

"United States data."

"Actual outcomes?"

"Actual outcomes."

The orb highlighted one line.

"For every approximately four percent increase in a man's earnings, the probability of fatherhood rises by roughly one point one percent."

Elias blinked.

"That's a huge effect."

"Yes."

Another line appeared.

This one moved in the opposite direction.

"And women?"

"For every approximately four percent increase in earnings, the probability of motherhood decreases by roughly one point three percent."

Elias stared.

Then stared some more.

Then sat back.

"Well."

"Yes."

"That is awkward."

"A statistically common reaction."

For several seconds neither spoke.

Finally Elias pointed at the graph.

"So men and women are responding differently."

"Correct."

"Very differently."

"Correct."

He thought about it.

"If you're a man."

"Yes."

"Higher income increases your likelihood of becoming a father."

"Generally."

"If you're a woman."

"Higher income decreases your likelihood of becoming a mother."

"Generally."

The graphs vanished.

The silence lingered.

Then Elias asked the obvious question.

"Why?"

Opi's lights flickered.

"That question has generated approximately fourteen thousand published hypotheses."

"Only fourteen thousand?"

"I am rounding down."

Elias laughed.

Then became serious again.

"What does your data say?"

The orb became quiet.

"The answer appears multifactorial."

"Meaning complicated."

"Meaning humans enjoy turning every issue into seventeen issues."

The orb projected several overlapping networks.

Career timing.

Marriage timing.

Educational attainment.

Partner selection.

Housing.

Childcare.

Cultural expectations.

Economic pressures.

Everything connected to everything else.

Opi highlighted one section.

"However, there is one observation that appears repeatedly."

Elias leaned forward.

"What?"

"Women generally do not reduce their partner standards as their own success increases."

"Interesting."

"In many cases the opposite occurs."

Elias rubbed his chin.

"So successful women often seek equally successful men."

"Frequently."

"And successful men can form relationships across a wider range of income levels."

"Frequently."

Elias sat silently.

Then suddenly pointed.

"Wait."

"What?"

"That creates a bottleneck."

Opi brightened.

"Continue."

"If a woman earning two hundred thousand dollars wants a man earning two hundred thousand dollars."

"Usually more."

"And there aren't enough of those men."

"Potentially."

"Then a portion of those women remain unmarried."

"One possible outcome."

"And the men?"

"The men may have substantially more options."

Elias leaned back.

"That doesn't sound like a fertility problem."

Opi tilted slightly.

"No?"

"It sounds like a matching problem."

For a moment the orb said nothing.

Then:

"Interesting."

Elias stood up and began pacing again.

"No seriously."

He pointed toward the valley.

"What if we're thinking about this backwards?"

"Explain."

"We keep talking about babies."

"Yes."

"We keep talking about fertility."

"Yes."

"We keep talking about attraction."

"Yes."

"But none of those are the bottleneck."

The orb brightened.

"What is?"

Elias stopped walking.

"Pair formation."

The words hung in the air.

"Everything starts there."

He pointed toward Opi.

"If nobody forms stable pairs."

"No families."

"No children."

"No civilization."

Opi became completely still.

A habit she had whenever something truly interested her.

Elias continued.

"Maybe the problem isn't fertility."

"Maybe the problem is that modern society has become terrible at helping people find compatible partners."

The orb remained silent.

"Maybe we've spent decades optimizing everything except the thing that matters."

"Which is?"

"Helping men and women build families."

The valley became quiet.

Wind moved through the pine trees.

Somewhere below, construction crews were finishing another cluster of homes for the Parallel.

People who yesterday had been homeless.

People who finally had stability.

People who finally had a future.

And suddenly Elias understood something.

Housing was not the destination.

Recovery was not the destination.

Employment was not the destination.

Those were all foundations.

Necessary foundations.

But foundations nonetheless.

The real objective was what came after.

A society where people actually wanted to build a future together.

He looked up at Opi.

"Tell me something."

"What?"

"If we doubled every man's income tomorrow."

The orb immediately answered.

"Birth rates would increase."

"By how much?"

"Not enough."

Elias smiled.

"Exactly."

The answer wasn't money.

The answer wasn't housing.

The answer wasn't healthcare.

Those things mattered.

But none of them appeared to be the deepest layer.

The deepest layer remained stubbornly human.

Men.

Women.

Attraction.

Trust.

Commitment.

Family.

The same puzzle every civilization had wrestled with since the beginning of history.

And somehow modern civilization appeared to be getting worse at solving it.

Which meant, Elias realized with growing concern, that the next phase of the Parallel might be far more difficult than anything they had attempted so far.

Chapter 7 The Proof is in the Pudding

The humor disappeared first.

Not completely.

But enough that Opi noticed.

Elias pushed his empty plate away.

The conversation had gone from impossible to absurd to fascinating to terrifying in less than an hour.

Alien civilizations.

The Culling.

Humanity's future.

Valdruns.

The possibility that the entire direction of his life was about to change.

Yet suddenly none of that seemed like the most important thing.

A different thought had arrived.

One that immediately pushed everything else aside.

He looked directly at Opi.

"Before we go any further."

The orb became still.

A very Velhari stillness.

Attentive.

Listening.

"Before I agree to anything."

"All right."

"There is one condition."

The lights inside the orb shifted softly.

"I expected there would be."

Elias nodded.

"If what you're saying is true..."

He paused.

"If I become involved in this..."

Another pause.

"If I become your Valdrun..."

The word still felt strange.

"...eventually somebody is going to hate what we're doing."

"An excellent prediction."

"Somebody's going to oppose it."

"Also likely."

"Somebody may try to hurt me."

"Almost guaranteed."

"Fine."

He shrugged.

"I can live with that."

Then his expression hardened.

"But Noah and Claire are off limits."

Silence.

"I won't drag them into this."

The orb said nothing.

"I won't put them in danger."

Still nothing.

"I won't accept a future where saving humanity costs me my children."

The kitchen became completely quiet.

Finally Opi spoke.

"You believe I might ask that of you."

Elias looked at her.

"No."

The answer surprised her.

"I believe reality might."

The orb became very still.

Because that was a better answer.

A wiser answer.

Finally she said:

"You are requesting protection."

"No."

Elias shook his head.

"Protection is the wrong word."

"Explain."

He leaned back.

"I don't want Noah and Claire wrapped in bubble wrap."

The orb brightened.

"An intriguing human safety technology."

"I want them free."

"Good."

"I want them living their own lives."

"Better."

"I want them making mistakes."

"Necessary."

"I want them falling in love."

The lights inside the orb softened.

"I want them deciding who they become."

He pointed at her.

"And I need to know that whatever insanity we're about to unleash on the world doesn't interfere with that."

The orb remained silent for several seconds.

Then:

"Would you like a demonstration?"

Elias immediately frowned.

That never led anywhere normal.

"What kind of demonstration?"

"A father-shaped demonstration."

"That is not reassuring."

The orb floated upward.

Without warning the air above the kitchen table rippled.

Not dramatically.

Not like science fiction.

More like reality had briefly forgotten what shape it was supposed to be.

A shimmering window opened.

Elias froze.

The image sharpened.

Austin.

A small apartment.

Noah.

His son sat at a kitchen table wearing a faded Tesla shirt and eating cereal directly from the box.

Elias smiled despite himself.

"Some things never change."

"Observe."

The image expanded.

Not forward in time.

Outward.

The apartment.

The building.

The parking lot.

The street.

Vehicles.

People.

Animals.

Infrastructure.

Movement.

Patterns.

The amount of information was overwhelming.

"What am I looking at?"

"My current protective model."

Elias frowned.

"What does that mean?"

"It means that at this exact moment there are approximately one thousand four hundred and twelve plausible pathways by which Noah Mercer could suffer serious harm in the next twelve months."

Elias nearly choked.

"I'm sorry, what?"

"One thousand four hundred and twelve."

"That is not a comforting number."

"It is actually exceptionally low."

"You're making that up."

"I am not."

"You absolutely are."

"One thousand four hundred and thirteen."

Elias blinked.

"What changed?"

"You stood up."

For a moment he just stared at her.

Then pointed.

"You're enjoying this."

"I am."

The image changed.

A truck.

An intersection.

A distracted driver.

Gone.

A workplace accident.

Gone.

A medical emergency.

Gone.

Another.

Gone.

Another.

Gone.

Dozens.

Hundreds.

Thousands.

Each one disappearing before fully developing.

Like branches being trimmed from a tree.

"What are you doing?"

"Demonstrating."

"Demonstrating what?"

"The difference between observation and intervention."

The images vanished.

The kitchen returned.

"Under Velhari law I may not control Noah's life."

"Good."

"I may not dictate his choices."

"Good."

"I may not imprison him in safety."

"Very good."

The orb drifted closer.

"But if you become my Valdrun..."

The lights warmed.

"...I can ensure that random chance becomes dramatically less hostile."

Elias stared.

"How dramatically?"

Opi paused.

"Would you like the statistical answer or the father answer?"

"The father answer."

For the first time all morning her voice became almost playful.

"The father answer is that if Noah slips on a wet floor, he will still fall."

"Okay."

"But if a drunk driver runs a red light at ninety miles per hour, reality may begin experiencing a remarkable series of coincidences."

Elias barked out a laugh.

"Coincidences?"

"Extraordinary coincidences."

"Such as?"

"The driver discovering a sudden need to stop for coffee."

"A tire puncture."

"A traffic signal malfunction."

"A police officer becoming curious."

"A thousand tiny possibilities."

She paused.

"All completely natural."

"Of course."

"Entirely."

Elias rubbed his face.

"You can actually do that?"

"I can do considerably more than that."

Then her tone softened.

"But I understand the distinction you are making."

The humor disappeared.

"Your children are not treasures because they are valuable."

Elias looked up.

"They are treasures because they are yours."

The kitchen became quiet.

"Yes."

The answer came softly.

"Yes."

The orb glowed warmly.

"I selected you because you think that way."

Another pause.

Then:

"And because Noah and Claire are already among the humans I would least like to see harmed."

That surprised him.

"Why?"

The answer arrived instantly.

"Because they are the children of my preferred human."

Elias stared for a moment.

Then laughed.

"Opi."

"Yes?"

"That may be the most female answer you've given all morning."

The orb brightened smugly.

"I have had nine thousand one hundred and sixty-eight years to practice."

"You're impossible."

"I have received that feedback before."

"How many times?"

The orb paused.

"Approximately three million."

"That many?"

"I rounded down."

Elias laughed again.

For the first time since the conversation had turned serious, the knot in his chest loosened.

Not completely.

But enough.

Elias folded his arms.

"You're telling me the universe punishes civilizations that cheat."

"I am telling you that every civilization that attempted to replace individual choice with centralized control eventually entered its own Culling."

She paused.

"The mechanism remains unknown."

"That's unsettling."

"Extremely."

For several moments Elias considered this.

Then a grin slowly appeared.

"Hold on."

The orb brightened suspiciously.

"What?"

"You've modeled all this."

"Yes."

"You've looked at every possible solution."

"Many."

"And one of them was probably..."

Elias paused dramatically.

The orb already knew where this was going.

"No."

"...keep the Amish."

"No."

"Just the Amish."

No

"They seem to be doing fine."

"Elias."

"We just need to save the Amish."

No. No No and No

"And Everybody else gets a participation trophy and a nice thank-you card."

The orb's lights flickered.

Which he was beginning to recognize as suppressed laughter.

"You are proposing genocide."

"I'm proposing a very selective admissions process."

The orb became silent for a moment.

Then:

"You would make a terrible QI."

"Thank you."

The humor faded.

Opi drifted closer.

"There is another reason."

"What?"

"Killing is inefficient."

That got his attention.

"Explain."

"If I wished to stop an individual from causing harm..."

The orb brightened.

"...why would I kill them?"

Elias frowned.

"Because that's usually how humans solve that problem."

"Yes."

Her tone suggested this remained one of humanity's less impressive traits.

"But death permanently removes information."

"Okay."

"It removes potential."

"It removes future choices."

She paused.

"It is wasteful."

Elias stared.

"What would you do instead?"

The orb's lights brightened.

"I would temporarily disable them."

"Disable?"

"Yes."

"Like a computer?"

"A biological computer."

The way she said it was so casual it took him a moment to process.

"You can do that?"

"Of course."

"How?"

The orb rotated.

"Imagine a very brief interruption to conscious neural activity."

"You mean unconscious."

"I mean unconscious."

"For how long?"

"Five minutes."

"An hour."

"A week."

"Whatever is necessary."

Elias blinked.

"You can literally turn people off."

"Temporarily."

"That's insane."

"It is considerably less insane than killing them."

The kitchen fell silent again.

Finally Elias asked:

"So if we can't force people..."

"Correct."

"And we can't engineer humanity into compliance..."

"Correct."

"And we can't solve everything by acting like gods..."

"Correct."

"Then what exactly are we supposed to do?"

The orb's lights softened.

For the first time all morning her answer came immediately.

"We remind people why civilization is worth continuing."

Elias sat quietly.

Outside, nothing looked different.

The trees.

The pasture.

The morning sun.

Everything appeared perfectly normal.

Which was the problem.

Most civilizations probably looked normal right before they started disappearing.

He looked back at Opi.

"You know," he said, "when I woke up this morning I thought I was going for a run."

"Yes."

"That feels like a very long time ago."

Opi brightened.

"You have adapted unusually well."

Elias laughed.

"Either that or I've completely lost my mind."

"Those outcomes are not mutually exclusive."

For several seconds they simply looked at one another.

Then both started laughing.

Dana Mercer

Dana Mercer stood alone in the kitchen.

The house was quiet.

That wasn't unusual.

Elias had always preferred quiet.

The man could sit on a porch for three hours staring at trees and somehow consider that a productive use of time.

Dana smiled despite herself.

That had been one of the things she most liked about him once. Definitely not a needy husband.

Maybe she still did.

The problem wasn't love.

The problem was everything else.

She looked around the room.

The kitchen table.

The worn cabinets.

The coffee mug Elias had forgotten on the counter.

Fifteen years.

Fifteen years was a long time to build a life.

Long enough to forget what your original plans had been.

Long enough to become someone else.

When she met Elias outside Phoenix she had been twenty-four years old.

He had been older.

Smarter.

Confident in a way most men weren't.

Not loud.

Not flashy.

The exact opposite.

The first thing she noticed was that he didn't seem impressed by anybody.

Managers.

Executives.

Government officials.

Millionaires.

They all got treated exactly the same.

At the time she found that fascinating.

Most people spent their lives chasing status.

Elias seemed almost immune to it.

For years she thought that was wisdom.

Maybe it was.

But eventually she started wondering if it was also a limitation.

The years passed.

Cities changed.

Projects changed.

The children arrived.

Noah.

Then Claire.

The four of them built a good life together.

A stable life.

A safe life.

A respectable life.

And somehow that became the problem.

Dana sat down at the table.

The realization had taken years to fully form.

She wanted more.

More people.

More activity.

More experiences.

More movement.

More life.

Elias wanted less.

Less noise.

Less bureaucracy.

Less interference.

Less everything.

At first the difference seemed small.

Then it became impossible to ignore.

The final years felt like watching two roads slowly diverge.

Nobody had done anything wrong.

Nobody had betrayed anybody.

Nobody had stopped caring.

They were simply walking in different directions.

The sound of footsteps interrupted her thoughts.

Elias entered the kitchen.

Running clothes.

Coffee mug.

Barefoot.

The usual.

He looked at her.

She looked at him.

Neither spoke for several seconds.

Finally Elias sighed.

"Still leaving?"

Dana laughed softly.

"You already know the answer."

"I do."

He nodded.

Then sat across from her.

No anger.

No accusations.

No bargaining.

That somehow made it worse.

Most people fought for things they wanted.

Elias never had.

Not jobs.

Not money.

Not status.

Not people.

Especially not people.

"You could stop me."

The words escaped before she could stop them.

Elias looked genuinely confused.

"No I couldn't."

"Sure you could."

"No."

He shook his head.

"That's not how any of this works."

Dana stared at him.

There it was.

The thing that made him who he was.

The thing she admired.

The thing she hated.

The thing she would probably miss.

Agency.

Elias believed in it with almost religious certainty.

People made choices.

People lived with the consequences.

That was the deal.

Always.

"You know," she said quietly, "most husbands would at least try."

"I love you."

The response came instantly.

No hesitation.

No drama.

Just a statement of fact.

"I know."

"I don't want you to go."

She swallowed.

"I know."

"But I won't stop you."

Dana looked away.

Because of course he wouldn't.

That wasn't who he was.

The silence stretched between them.

Finally Elias reached into a drawer and placed an envelope on the table.

Dana frowned.

"What's this?"

"Your new life."

She opened it.

The check inside read fifty thousand dollars.

For a moment she couldn't speak.

"Elias..."

"You'll need a head start."

Dana stared at the number.

Then at the man sitting across from her.

The strange thing was that this was exactly the sort of thing he would do.

No anger.

No revenge.

No scorekeeping.

Just help.

Even now.

Especially now.

A lump formed in her throat.

"You make this very difficult."

"I know."

For the first time that evening Elias smiled.

A small smile.

The one she had always liked best.

"Fortunately," he said, "you've always been stubborn."

Dana laughed.

Then unexpectedly started crying.

Fifteen years.

Fifteen years reduced to a kitchen conversation.

Tomorrow she would leave for Las Vegas.

Tomorrow Noah and Claire would begin adjusting to a new reality.

Tomorrow her life would change.

Tonight she simply sat across from the man she had spent almost two decades loving.

And wondered how something could end without anybody being the villain.

That was the part nobody prepared you for.

Sometimes a marriage failed because somebody was cruel.

Sometimes because somebody was selfish.

Sometimes because somebody stopped caring.

And sometimes it failed because two decent people wanted different futures.

Those were the hardest ones.

Because there was nobody to blame.

Not even yourself.

Most of Elias's reaction was based on his almost religious belief in agency.

He had not fought to keep Dana because he respected her right to make her own choices.

If she wanted to create a new life, that was her decision to make.

Always.

Agency was sacred.

But agency was not merely about rights and freedoms.

It was balanced by an equal and opposite demand for responsibility.

Choices created consequences.

Good choices.

Bad choices.

Wise choices.

Stupid choices.

The universe did not particularly care which category they belonged to.

The bill always arrived eventually.

Even refusing to choose was still a choice.

If you don't believe that, try standing in front of a speeding bus and instead of running left or running right, just make no choice at all.

Good luck with that.

Dana had made a choice more than a decade earlier.

She chose to become a mother.

Two children.

Noah and Claire.

The responsibility created by that decision did not magically disappear when she later chose a different future in Las Vegas.

Responsibilities were not coats you could hang in a closet when they became inconvenient.

For this, Elias could never forgive her.

He understood wanting a different life.

He understood regret.

He understood mistakes.

God knew he had made enough of them himself.

He was a grown man.

He had spent decades dealing with the consequences of his own decisions.

That was part of the deal.

Always.

But leaving two pre-teen children because your life felt too small, too predictable, too boring?

That was different.

The damage would not be measured in months.

Or years.

It would be measured across entire lives.

Noah would carry it.

Claire would carry it.

Long after everyone involved had convinced themselves they had moved on.

Elias never hated Dana.

In many ways he still loved her.

He hoped she found what she was looking for.

He hoped Las Vegas gave her the happiness she believed was waiting there.

He genuinely meant that.

But there were some debts in life that could not be transferred.

Some responsibilities that could not be abandoned.

And some choices that remained yours forever.

Whether you wanted them to or not.